Shortcut Audit, Causal Bands, and an Out-of-Distribution Recipe for Vibration-Based Water Leak Detection

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Abstract

A 99 % accuracy reported on the Korean AI Hub water-pipe leak benchmark collapses to weighted accuracy 56.8 % and macro-F1 0.304 under realistic deployment shift — and a four-rule lookup that ignores the entire vibration spectrum already reaches 99.77 %. Tabular machine-learning pipelines for water-pipe leak detection on vibration data routinely report such row-level accuracies, yet they rarely survive field deployment. The AI contribution of this paper is a reusable methodology for auditing identifier shortcuts and quantifying out-of-distribution (OOD) brittleness in any sensor-ML benchmark; the engineering contribution is a deployment-candidate leak-typing recipe for municipal water-utility networks. The methodology is instantiated on the Korean AI Hub vibration benchmark (70,384 recordings, 49 sites, 12 administrative regions) used in the author's prior study (Lee & Kim 2023). Four findings. (i) A four-probe shortcut audit: a rule-based identifier lookup that ignores the entire vibration spectrum reproduces 99.77 % validation accuracy, within 0.02 percentage points of the published state of the art. (ii) A three-tier OOD ladder (row-CV → GroupKFold-by-site → Leave-One-Region-Out) plus a 16-fold county-level ablation: under Leave-One-Region-Out, XGBoost falls from 94.0 % to weighted 56.8 % accuracy and macro-F1 0.304. (iii) Three independent attribution methods converge on **121–422 Hz** as the robustly generalisable leak band (termed "causal" in the operational OOD sense throughout this paper), with 422–1470 Hz exposed as an active confounder whose removal *improves* OOD macro-F1; the finding is validated against seven additional baselines, including gradient-boosted ensembles, attentive tabular deep learning, and domain-alignment methods (correlation alignment, domain-adversarial training, and maximum-mean-discrepancy generalisation). (iv) A multi-seed deployment recipe: a tree-only soft average of flat and hierarchical XGBoost on 57 causal features attains OOD macro-F1 **0.4326** with perfect normal-class recall; the strongest deep alternative (an MLP trained with group distributionally robust optimisation, GroupDRO) reaches only 0.358 ± 0.013 across five seeds. The audit protocol is independently re-applied to nine external public sensor benchmarks (UCI SECOM McCann and Johnston (2008), UCI HAR Anguita et al. (2013), and the seven machines of MIMII-DG Dohi et al. (2022)) and shown to discriminate shortcut-laden from clean benchmarks rather than blanket-flagging vibration data. The full audit protocol, multi-seed harness, and deployment recipe are released for reproducibility.

Keywords: water leak detection, vibration sensor, out-of-distribution evaluation, shortcut learning, causal feature selection, smart-water-grid

1. Introduction

Non-revenue water loss from leaking metal and non-metal distribution pipes is a persistent infrastructure problem with substantial economic and environmental cost worldwide, particularly acute in aging urban networks. Acoustic leak detection using low-cost vibration sensors mounted at water meters, valves, and hydrants has emerged as one of the most widely deployed monitoring strategies because of its low installation cost, non-invasiveness, and compatibility with cellular telemetry Hunaidi et al. (2000); Ismail et al. (2019). On the sensing side, the dominant trend

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over the past decade has been to replace hand-tuned threshold rules with tabular machine learning (ML) applied to frequency-domain features of the vibration spectrum Shukla and Piratla (2020); Fereidooni et al. (2021); Cody et al. (2020); Liu et al. (2019).

The study that motivates the present follow-up is the author's own previous work on the Korean AI Hub water-pipeline benchmark Lee and Kim (2023), in which eight tabular models were compared across four feature regimes and an XGBoost classifier reached 99.79 % accuracy under stratified 10-fold cross-validation. The goal of that paper was a comparative benchmark of off-the-shelf models on newly curated sensor data, and within its stated evaluation protocol the result is faithful. However, on subsequently attempting to operationalise the same pipeline for field use by a municipal water utility, two practical obstacles became visible. First, the 10-fold row-level split placed multiple recordings from the same physical sensor in both train and test folds, so the reported accuracy actually measured within-sensor repeatability more than cross-sensor generalisation. Second, as the shortcut-learning literature in medical imaging Zech et al. (2018); DeGrave et al. (2021); Geirhos et al. (2020); D'Amour et al. (2022) and natural-image recognition Beery et al. (2018) has documented, sensor-ML models routinely latch onto identifier-like features that correlate perfectly with the label in the training distribution but do not represent any physical signal.

Scope of this paper:. This paper addresses that gap with a reusable AI methodology and a deployment-candidate recipe. Treating the AI Hub benchmark as one carefully studied case rather than the exclusive object of study, four contributions of broader interest to engineering AI practitioners are made.

1. **A reusable four-probe shortcut-audit protocol for sensor-ML benchmarks.** Per-sensor label purity, class-lookup reconstruction, sensor train/test overlap, and a meta-column-override probe. On AI Hub, a four-rule identifier lookup that ignores the entire vibration spectrum reproduces 99.77 % validation accuracy — within 0.02 percentage points of the published state of the art Lee and Kim (2023).
2. **A three-tier out-of-distribution evaluation protocol with a county-level extension.** Row-CV → GroupKFold-by-site → Leave-One-Region-Out (LORO) over 12 *gwangyeoksi-do*, plus a 16-fold Leave-One-County-Out over the eligible *sigungu* (16 of 36 with ≥ 4 leak classes present). Under LORO the XGBoost classifier of Lee and Kim (2023) falls to weighted accuracy 56.8 % and macro-F1 0.304.
3. **Identification of the robustly generalisable spectral signature of leak noise.** Three independent attribution methods — bootstrap permutation importance, TreeSHAP Lundberg and Lee (2017), and multi-region replication — converge on the **121–422 Hz** band, broadly consistent with published plastic-pipe acoustics Hunaidi et al. (2000); Almeida et al. (2014); Gao et al. (2017). The 422–1470 Hz range is exposed as an active confounder whose permutation *improves* OOD macro-F1. Following Arjovsky et al. (2019); Schölkopf et al. (2021), the selected band is called "causal" in the operational sense of "necessary for OOD generalisation" throughout this paper (Remark 1).
4. **A multi-seed deployment recipe with a full computational-cost analysis.** Restricting the input to 57 causal features and combining a flat causal XGBoost with a hierarchical (binary energy gate + 4-class causal typer) via a pre-specified soft average attains OOD macro-F1 **0.4326** with perfect normal-class recall on the canonical LORO fold. The recipe is compared on macro-F1 against seven additional baselines (LightGBM Ke et al. (2017), CatBoost Prokhorenkova et al. (2018), TabNet Arik and Pfister (2021), Random Forest Breiman (2001), DeepCORAL Sun and Saenko (2016), DANN Ganin et al. (2016), MMD-DG Li et al. (2018)); training time, inference latency per 1,000 rows, peak RAM and peak GPU memory are reported for the deployment-relevant subset (XGBoost flat, XGBoost hierarchical, tree-ensemble, LightGBM, CatBoost, Random Forest, and the multi-layer perceptron (MLP) / one-dimensional convolutional neural network (1D-CNN) backbones under empirical risk minimisation (ERM)).

Figure 1 summarises the four stages of the study as a single pipeline diagram, anchoring each stage to the headline quantitative result that section will defend.

The remainder of the paper is organised as follows. Section 2 surveys the four bodies of literature this work intersects (vibration leak detection, shortcut learning, OOD generalisation, tabular deep learning). Section 3 formalises the methodology with four definitions and an applicability checklist. Section 4 describes the dataset, multi-seed protocol, hardware, and reproducibility infrastructure. Section 5 presents the experimental results organised around seven research questions. Section 6 discusses engineering implications and translates the recipe into operational metrics,

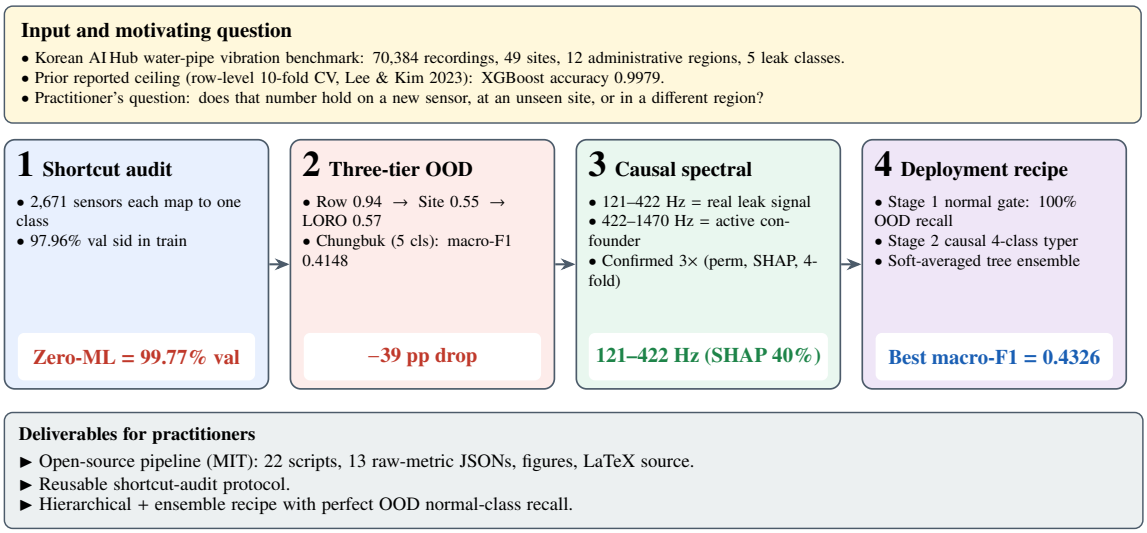


Figure 1: **Overview of the study.** The paper proceeds as a four-stage pipeline from the benchmark ceiling reported in Lee and Kim (2023) to a deployment-candidate recipe. (1) A zero-ML identifier lookup reproduces the reported 99.77 % accuracy, so the signal contribution to the headline number is negligible. (2) A three-tier OOD evaluation exposes a ~39-percentage-point accuracy drop between row-level and GroupSite-by-site cross-validation, and a further drop to weighted accuracy 56.8 % under Leave-One-Region-Out. (3) Three independent attribution methods (permutation importance, TreeSHAP, cross-region bootstrap) converge on **121–422 Hz** as the generalisable leak-signal band, with 422–1470 Hz identified as an active confounder. (4) A pre-specified soft-probability average of flat and hierarchical XGBoost on the 57 causal features attains OOD macro-F1 **0.4326** with perfect normal-class recall on the canonical Chungbuk fold.

a cost-benefit estimate, and a SCADA/IoT integration template. Section 7 catalogues threats to validity and failure modes. Section 8 concludes.

2. Related Work

2.1. Vibration-Based Leak Detection

Acoustic signatures of water leaks in buried pipes have been studied since at least the late 1990s. For plastic pipes, Hunaidi et al. (2000) reported that leak-noise energy is concentrated below roughly 600 Hz, with the precise band depending on pipe material, diameter, and internal pressure. Almeida et al. (2014) quantified the acoustic filtering introduced by sensor mounting and pipe-wall damping in buried plastic pipes; Gao et al. (2017) improved the shape of the cross-correlation function for time-delay-based leak detection in plastic distribution pipes by removing pipe-resonance effects, broadening the usable bandwidth of the analysis. Ismail et al. (2019) systematically reviewed accelerometer-based approaches, classifying the literature into threshold methods, signal-processing methods (cross-correlation, EMD, wavelet), and machine-learning methods.

Recent representative ML works include Shukla and Piratla (2020) (supervised classification of acceleration signals), Fereidooni et al. (2021) (hybrid model-based methods for large distribution networks), Cody et al. (2020) (deep autoencoders on hydroacoustic spectrograms), Liu et al. (2019) (machine-learning over wireless-sensor-network signals), and Tariq et al. (2022) (data-driven leak detection with MEMS-based accelerometers). Korean studies on the AI Hub benchmark Lee and Kim (2023) have reported above-99 % accuracy with tree ensembles such as XGBoost Chen and Guestrin (2016) and Random Forest Breiman (2001). The reported numbers are measured under stratified row-level cross-validation on a merged-and- resplit training set; no published evaluation on the benchmark has used a sensor-group, site-group, or region-group holdout, and none has audited for identifier shortcuts. This paper supplies that evaluation.

2.2. Shortcut Learning and Spurious Correlation

The shortcut-learning framework Geirhos et al. (2020) provides a vocabulary for the failure mode in which a supervised model latches onto superficial features that correlate with the label in training but do not represent the

underlying task. Zech et al. (2018) showed that a deep pneumonia classifier was effectively detecting the hospital of origin via metadata artefacts in radiographs; DeGrave et al. (2021) documented analogous shortcut behaviour in COVID-19 chest-X-ray classifiers; Beery et al. (2018) demonstrated that animal-species recognisers trained on wildlife camera imagery rely heavily on background context. D'Amour et al. (2022) frame the phenomenon more generally: many equally accurate models on the training distribution can behave very differently under deployment shift. Liu et al. (2021) propose Just-Train-Twice as a group-information-free worst-group remedy. Formal shortcut audits have been rare in the sensor-ML literature; this paper adapts the protocol to a vibration-based leak-detection benchmark and finds a particularly extreme form of the pathology.

2.3. Out-of-Distribution Generalisation and Domain Alignment

Loss-level invariance regularisers are the canonical algorithmic response to spurious correlations. IRM Arjovsky et al. (2019) adds a penalty that discourages representations whose risk varies across training environments; GroupDRO Sagawa et al. (2020) minimises the worst-case per-environment loss; REx Krueger et al. (2021) penalises the variance of per-environment risks; Fishr Rame et al. (2022) operates on gradient variances. The empirical reliability of these methods has been repeatedly questioned Gulrajani and Lopez-Paz (2021); Rosenfeld et al. (2021): with few or uneven environments they often fail to beat plain Empirical Risk Minimisation on OOD test sets.

A complementary line of work — domain-adaptation and domain-generalisation methods — aligns feature distributions between source and target environments. DeepCORAL minimises the Frobenius distance between source and target covariance matrices Sun and Saenko (2016); DANN trains a domain classifier through a gradient-reversal layer Ganin et al. (2016); MMD-DG Long et al. (2015); Li et al. (2018); Gretton et al. (2012) minimises the maximum mean discrepancy across environments. Section 5 contributes evaluations of all three on the AI Hub LORO fold.

2.4. Tabular Deep Learning

Tabular deep learning has matured rapidly with the introduction of TabNet Arik and Pfister (2021), NODE Popov et al. (2020), and FT-Transformer Gorishniy et al. (2021). The recent survey of Borisov et al. (2024) reports that, on most benchmarks, gradient-boosted decision trees (XGBoost Chen and Guestrin (2016), LightGBM Ke et al. (2017), CatBoost Prokhorenkova et al. (2018)) remain competitive with — and frequently exceed — the strongest deep tabular architectures, particularly on small to medium datasets. All three boosting variants and TabNet are included as additional baselines on the AI Hub fold.

2.5. Reproducibility and Industrial AI

Concern about empirical robustness in ML benchmarking has driven the development of reproducibility surveys (Pineau et al. Pineau et al. (2021); Henderson Henderson et al. (2018)) and the WILDS distribution-shift benchmark Koh et al. (2021). AI risk frameworks for high-stakes infrastructure — the NIST AI RMF National Institute of Standards and Technology (2023) and the EU AI Act European Parliament and Council (2024) high-risk categorisation of critical infrastructure — make explicit deployment-evaluation requirements that the shortcut-audit protocol of this paper directly supports.

3. Problem Formulation and Methodology

3.1. Problem Setting and Notation

Let $\mathcal{D} = \{(x_i, y_i, s_i, e_i)\}_{i=1}^{N}$ denote a sensor-ML benchmark, where $x_i \in \mathbb{R}^d$ is the feature vector, $y_i \in \{1, \dots, C\}$ is the class label, s_i is a per-sensor identifier, and $e_i \in \{1, \dots, E\}$ is a higher-order group identifier (site, region). Let ℓ denote a class-weighted softmax cross-entropy with inverse-frequency class weights $w_c = N/(CN_c)$, $R(\theta) = N^{-1} \sum_i \ell(f_\theta(x_i), y_i)$ the pooled risk, and $R_e(\theta) = N_e^{-1} \sum_{i:e_i=e} \ell(f_\theta(x_i), y_i)$ the per-environment risk. The feature vector is assumed to partition into G disjoint physical groups $x = (x^{(1)}, \dots, x^{(G)})$ (e.g. frequency bands, multivariate channels, segments of a trajectory).

3.2. Definition 1: Identifier-Shortcut Sensor Benchmark

Definition 1 (Identifier-shortcut benchmark)**.** *A sensor-ML benchmark $\mathcal{D}$ together with a row-level train/validation split admits an* identifier shortcut *of strength (ϵ, δ) if there exists a deterministic mapping $h\colon (s, m) \mapsto y$, depending only on the identifier s and any meta-columns m (timestamp, sample-rate flags, etc.) but not on x, such that*

$$\frac{1}{N_{\text{val}}} \sum_{i \in \text{val}} \mathbb{1}[h(s_i, m_i) = y_i] \;\geq\; 1 - \epsilon,$$

and $\Pr[s_i \in \text{train} \mid i \in \text{val}] \;\geq\; 1 - \delta$.

A benchmark with $(\epsilon, \delta) \approx (0, 0)$ contains a near-perfect identifier shortcut: any reported validation accuracy is bounded above $1 - \epsilon$ by h alone, so the contribution of any model trained on x is at most ϵ in absolute accuracy. Section 5 demonstrates that the AI Hub benchmark with the standard split satisfies $(\epsilon, \delta) \leq (0.0023, 0.0204)$, which means the published 99.79 % accuracy Lee and Kim (2023) is within rounding of the lookup ceiling.

3.3. Definition 2: Four-Probe Shortcut Audit

Definition 2 (Four-probe shortcut audit)**.** *Given $\mathcal{D}$ and a row-level train/val split, the* four-probe shortcut audit *computes:*

- ***P1 (per-sensor label purity)****: for each training sensor s, the fraction of rows at s whose label matches the modal label at s. Median purity ≥ 0.95 flags single-class sensors.*
- ***P2 (class-lookup reconstruction)****: build $\widehat{h}_{\text{lookup}}(s) := \arg\max_y \Pr_{\text{train}}(y \mid s)$ and apply it to validation rows. Report the coverage and covered-row accuracy.*
- ***P3 (sensor train/test overlap)****: the fraction of validation sensors whose identifier s also appears in the training split.*
- ***P4 (lookup + meta-column override)****: combine P2 with a rule-based override on any meta-column whose value is deterministically associated with a specific class. Report the resulting accuracy.*

A benchmark passes the audit if no probe reaches the published benchmark accuracy; it fails if P1, P3, and P4 jointly do.

3.4. Definition 3: Three-Tier OOD Evaluation Ladder

Definition 3 (Three-tier OOD evaluation)**.** *Evaluation under increasing deployment realism:*

- ***Row-CV.*** *Stratified k-fold over rows. The same sensor's measurements may appear in both folds.*
- ***GroupSite-CV.*** *GroupKFold by* `site`*, so every test sensor has been unseen at train time. Simulates deployment to a new installation within the same region.*
- ***LORO.*** *Leave-One-Region-Out over the administrative regions. Simulates deployment to a new city or province.*

The gap between row-CV and LORO on the same model gives a quantitative measure of the deployment-shift brittleness of a pipeline.

A finer-grained Leave-One-County-Out (LOCO) protocol is also evaluated on the AI Hub benchmark, drawing held-out test folds from the 36 unique *sigungu* (county) codes — restricted to the 16 eligible folds with at least four leak classes present — versus the 12 environments at the *gwangyeoksi-do* (metropolitan-region) level. LOCO tests whether more environments stabilise gradient-based invariance regularisers (IRM Arjovsky et al. (2019), GroupDRO Sagawa et al. (2020)).

3.5. Definition 4: Grouped Permutation Importance

Definition 4 (Grouped permutation importance)**.** *After training f_θ on the pooled training set, the causal contribution of a feature group $g \in \{1, \ldots, G\}$ is measured on the held-out set by*

$$\Delta_{\mathrm{F1}}(g) = \mathrm{F1}(y, f_\theta(x)) - \mathrm{F1}(y, f_\theta(\tilde{x}^{(g)})),$$

where $\tilde{x}^{(g)}$ permutes only the columns indexed by group g. Positive $\Delta_{\mathrm{F1}}(g)$ marks g as carrying generalisable signal; negative $\Delta_{\mathrm{F1}}(g)$ marks g as a confounder whose removal helps. A bootstrap confidence interval on $\Delta_{\mathrm{F1}}(g)$ is obtained by B independent permutations of $x^{(g)}$.

Definition 4 is further paired with a SHAP second-source attribution Lundberg and Lee (2017), declaring a group g *robustly generalisable* on the test fold only if $\Delta_{\mathrm{F1}}(g) > 0$ *and* the SHAP-magnitude share of g is in the top quartile. SHAP is one of several feature-attribution methods that could be substituted at this step; LIME Ribeiro et al. (2016) offers a local-surrogate alternative, and causal Shapley values Janzing et al. (2020) provide a stronger interpretation when the data-generating process is partly known Lipton (2018); Barredo Arrieta et al. (2020). TreeSHAP is used here because the underlying model is a gradient-boosted tree ensemble for which exact path-based attribution is tractable. Causal Shapley values were not adopted because they require a known or estimated causal graph among features, which is unavailable for the 513 frequency bins: while the bins are physically ordered, their conditional independence structure is not known a priori and would need to be estimated from data — introducing a circularity that the simpler permutation-plus-SHAP protocol avoids.

Remark 1 (Use of the term "causal"). Throughout this paper, a feature group is called *causal* in the operational sense that (i) its permutation degrades OOD macro-F1 on held-out regions, (ii) two independent attribution methods (permutation importance and TreeSHAP) converge on the same group, and (iii) the identified band (121–422 Hz) is physically consistent with published plastic-pipe leak acoustics Hunaidi et al. (2000); Almeida et al. (2014); Gao et al. (2017). This usage follows the "necessary for OOD generalisation" convention common in the distribution-shift literature Arjovsky et al. (2019); Schölkopf et al. (2021) and does *not* imply a formal interventionist causal claim in the structural-causal-model sense of Pearl (2009). A randomised field experiment — physically masking specific frequency bands at recording time — would be required for such a claim and is identified as future work.

3.6. Theoretical Framing

The relationship between the identifier-shortcut audit and the three-tier OOD ladder admits a concise lower bound that formalises why shortcuts are catastrophic under group cross-validation.

Proposition 1 (Row-CV → Group-CV accuracy drop)**.** *Let $\mathcal{D}$ admit an identifier shortcut of strength (ϵ, δ) in the sense of Definition 1, and let f_θ be a classifier trained by empirical risk minimisation on the row-level split. Assume* (i) *the row-level validation split has identifier overlap rate $\geq 1 - \delta$ with the training split,* (ii) *each identifier s has per-sensor label purity 1 in the training split (single class per sensor), and* (iii) *under group cross-validation the held-out identifiers s are unseen at train time. Then*

- *the row-CV accuracy of f_θ is at least $(1 - \epsilon - \delta)$, achievable purely by an identifier-conditional posterior $\Pr_{\mathrm{train}}(y \mid s)$;*
- *the group-CV accuracy of any classifier whose prediction depends only on s collapses to the marginal-class baseline $\rho := \max_y \Pr(y)$.*

Consequently, the row-CV → group-CV accuracy drop attributable to identifier shortcuts is at least $1 - \epsilon - \delta - \rho$.

Proof sketch.. Under (*i*+*ii*), the identifier lookup $\widehat{h}(s) = \arg\max_y \Pr_{\mathrm{train}}(y \mid s)$ is correct on every row whose identifier appears in the training split, which is $\geq 1 - \delta$ of validation rows; the remaining δ fraction contributes at most ϵ error, giving the $1 - \epsilon - \delta$ floor. Under (*iii*), s at test time is independent of the training posterior, so any prediction that depends only on s necessarily reverts to the marginal ρ. Subtracting gives the stated bound. □

Practical implication.. For the AI Hub benchmark, the empirical instantiation of Proposition 1 gives $(\epsilon, \delta, \rho) \leq (0.0023, 0.0204, 0.315)$, predicting a row-CV → group-CV drop of at least 0.66 accuracy points attributable to shortcut alone. The measured drop in Section 5.2 (Table 3, $0.945 \rightarrow 0.553 = -0.39$) is smaller than this upper-bound prediction, which is exactly what one expects: the residual 0.27 gap between predicted (0.66) and observed (0.39) is the residual physical signal that x still carries about y once identifiers are removed.

Tightness of the bound.. Proposition 1 is intentionally loose: it lower-bounds the *minimum possible* drop when all prediction relies on identifiers, whereas a real model uses both identifiers and signal. The bound is tight in the degenerate case where f_θ learns a pure identifier lookup (then the entire row-CV accuracy is attributable to shortcut and the full $1 - \epsilon - \delta - \rho$ drop is realised). For models that also learn signal features, the observed drop is smaller than the bound by exactly the amount of physical signal that survives. The bound's practical value is therefore as a *diagnostic ceiling*: if the observed drop is close to $1 - \epsilon - \delta - \rho$, the model learned almost nothing beyond the shortcut; if the drop is much smaller, there is recoverable signal in x. On AI Hub, the ratio of observed (0.39) to predicted (0.66) suggests roughly 41 % of the row-CV accuracy is signal-based, providing a quantitative budget for causal feature selection. Causal feature selection — restricting x to a physically nameable subset $x^{(g^*)}$ — targets that residual signal at the input, an approach aligned with recent arguments for causal representation learning Schölkopf et al. (2021). Loss-level invariance regularisers (IRM, GroupDRO) attempt to penalise reliance on s during training but inherit the data-side problem if the training environments have heavily overlapping identifiers.

3.7. Causal Feature Selection vs. Loss-Level Invariance

Two algorithmic families are instantiated on the same MLP and CNN backbones for fair comparison. Loss-level methods comprise ERM, IRMv1 Arjovsky et al. (2019) (with $\lambda_{\max} \in \{1, 10, 10^2, 10^3, 10^4\}$ and $T_{\text{warmup}} \in \{10, 30\}$), and GroupDRO Sagawa et al. (2020) ($\eta = 10^{-2}$). The domain-alignment family adds DeepCORAL Sun and Saenko (2016), DANN Ganin et al. (2016), and a multi-bandwidth MMD-DG Long et al. (2015); Li et al. (2018). The causal-feature family uses the 65–422 Hz subset selected by the grouped permutation importance of Definition 4 and runs on XGBoost, LightGBM, CatBoost, TabNet, and Random Forest.

3.8. Hierarchical Two-Stage Classifier

For deployment, the 5-class problem factorises into a binary gate and a 4-class typer. Stage 1 is a binary XGBoost trained on the 57 causal features with target $\mathbb{1}[y = \texttt{normal}]$. Stage 2 is a 4-class XGBoost trained on the abnormal subset; at inference, Stage 1 routes samples and Stage 2 is consulted only when Stage 1 predicts "abnormal". The Stage 1 threshold is swept over $T \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ to characterise the missed-leak / false-alarm tradeoff (Section 5).

3.9. Permutation-Guided Attention (Negative-Result Ablation)

As a novelty ablation, an MLP is augmented with a learnable per-band attention vector $a \in \mathbb{R}^{513}$ whose softmax is regularised toward a prior p derived from permutation importance computed on an internal 80/20 train-only split (no test-set leakage):

$$\mathcal{L}_{\text{PGA}}(\theta) = R(\theta;\ x \odot \text{softmax}(a)) + \lambda_c\, \text{KL}(\text{softmax}(a) \,\|\, p).$$

Because the prior is itself computed on training data with near-complete sensor overlap, it inherits the shortcut bias it was meant to regularise away — making this a soft-prior counterpart to hard causal feature selection. Whether it recovers the causal-XGBoost ceiling is reported as a negative result.

3.10. Soft-Probability Ensemble

To test whether combining the strongest tree models offers additional headroom, a soft-probability ensemble averages the flat causal XGBoost and the hierarchical classifier. The deployment candidate is fixed in advance as this tree-only average, ensuring the reported number is not selected on the test fold.

3.11. Applicability Checklist

The methodology of Sections 3.3–3.10 applies to any sensor-ML benchmark with: (i) at least one identifier-like column (sensor ID, patient ID, device ID), (ii) a natural hierarchy of group variables (sensor / site / region / time), (iii) features that group physically (frequency bands, channels, segments), and (iv) a published row-level CV result that the audit can target. Examples include gas-pipe leak detection, rotating-machinery condition monitoring, partial-discharge detection in power transformers, and sound-based machine-fault diagnosis.

4. Dataset, Multi-Seed Protocol, and Reproducibility

4.1. AI Hub Vibration Benchmark

The AI Hub water-pipe leak vibration dataset NIA Korea AI Hub (2022); Lee and Kim (2023) consists of $N = 70{,}384$ time-frequency feature vectors extracted from vibration recordings at 2,710 unique sensor identifiers across 49 installation sites in 12 *gwangyeoksi-do* (metropolitan-region) codes and 36 unique *sigungu* (county) codes. Each recording is represented as a 513-component spectrum covering 0–5120 Hz at 10 Hz resolution (the 0 Hz column is identically zero), 20 peak-amplitude aggregates `MAX0`, . . . , `MAX19`, and metadata columns `site`, `sid`, `ldate`, `lrate`, `llevel`, and `leaktype` $\in$ {`in`, `out`, `noise`, `other`, `normal`}. The class distribution is 31.5 % `normal`, 28.0 % `out`, 21.2 % `in`, 11.2 % `other`, 8.0 % `noise`. The dataset ships with a 62,564 / 7,820 train/val split.

The 12 administrative regions are unevenly populated: three (Jeonnam, Jeonbuk, Gyeongbuk) account for 69 % of the rows; five contain fewer than 1,000 rows each. Only Chungbuk (region 43, 5,074 rows) contains substantial counts of all five leak classes. Chungbuk therefore serves as the canonical 5-class OOD test set.

4.2. Multi-Seed Protocol

Every stochastic experiment in this paper is run across five seeds $S = \{0, 1, 2, 3, 42\}$, with SEED = 42 matching the original configuration of Lee and Kim (2023) and the four additional seeds providing variance estimates. The data partition is held fixed (data-split seed = 42) so that the only varying source between seeds is model initialisation and mini-batch ordering. Mean ± standard deviation over the five seeds is reported for every headline number, accompanied by paired Wilcoxon tests for method comparisons.

4.3. Hardware and Reproducibility

XGBoost / LightGBM / CatBoost / Random Forest experiments run on a 16-core CPU. Deep-learning experiments use two NVIDIA TITAN Xp GPUs (12 GB each, CUDA 12.2 driver 535) with one GPU per backbone. Total compute for the full multi-seed sweep is approximately 30 h wall-clock. Code, multi-seed runner, result JSONs, and figure-generation scripts are released under the MIT licence at `https://github.com/suanlab/causal-leak` (see Data Availability). The `run_multiseed.sh` script reproduces every figure and table from the paper end-to-end via a single command.

Table 1 lists the hyperparameters for every model family evaluated in this paper. All settings are fixed before the test fold is examined; no hyperparameter was selected on the Chungbuk LORO test set.

5. Results

The results are organised around seven research questions (RQ1–RQ7), each anchored by a figure or table.

5.1. RQ1 — Does the published 99.79 % benchmark survive an identifier shortcut audit?

Applying the four probes of Definition 2 to the AI Hub dataset:

P1 (per-sensor label purity): all 2,671 training sensors have purity 1.0. Each `sid` maps to exactly one leak class for its entire lifetime, a consequence of the per-class file structure in the AI Hub release.

P3 (train/test overlap): 1,876 of 1,915 validation sensors (97.96 %) also appear in training.

P2 (class lookup): an `sid` $\rightarrow$ majority-label lookup covers 99.5 % of the validation set with 100 % accuracy on covered rows.

P4 (lookup + `lrate` override): combining the `sid` lookup with the rule "`lrate` $\neq$ 90 $\Rightarrow$ predict `normal`" yields 99.77 % validation accuracy — within 0.02 percentage points of the 99.79 % reported in Lee and Kim (2023).

The benchmark fails Definition 2: its $(\epsilon, \delta) \approx (0.0023, 0.0204)$, i.e. a near-perfect identifier shortcut.

Table 1: Hyperparameter settings for all model families. Tree models share the same 200-estimator, depth-6, lr=0.1 budget for fair comparison. Deep models use the architecture and schedule shown; only the seed varies across the five-seed protocol.

Model	Family	Key hyperparameters
XGBoost	tree	`n_estimators`=200, `max_depth`=6, `learning_rate`=0.1, `tree_method`=hist, `objective`=multi:softprob, inverse-frequency class weights
LightGBM	tree	`n_estimators`=200, `num_leaves`=63, `max_depth`=6, `learning_rate`=0.1, `objective`=multiclass, inverse-frequency sample weights
CatBoost	tree	`iterations`=200, `depth`=6, `learning_rate`=0.1, `loss_function`=MultiClass, inverse-frequency class weights
Random Forest	tree	`n_estimators`=400, `max_depth`=20, `class_weight`=balanced
MLP	deep	57→256→128→5, `epochs`=120, `batch_size`=528, `lr`=10^{-3}, `weight_decay`=10^{-4}, early-stop patience=25
1D-CNN	deep	513-band input, 3×(Conv1d→BN→ReLU→Pool), fc head 128→5, same schedule as MLP
TabNet	tab. DL	`n_d`=`n_a`=32, `n_steps`=4, γ=1.5, λ_{sparse}=10^{-4}, `max_epochs`=80, `patience`=15, `batch_size`=2048, CPU only
IRM (MLP)	invariance	Same MLP backbone; $\lambda_{\text{max}} \in \{1, 10, 10^2, 10^3, 10^4\}$, $T_{\text{warmup}} \in \{10, 30\}$; best: λ=100, T=30
GroupDRO (MLP)	invariance	Same MLP backbone; η_{dro}=10^{-2}
DeepCORAL	domain-align	MLP backbone; λ_{coral}=1.0
DANN	domain-align	MLP backbone; α_{max}=1.0, λ_{disc}=0.5, gradient-reversal layer
MMD-DG	domain-align	MLP backbone; λ_{mmd}=1.0, bandwidths={0.01, 0.1, 1, 10, 100}

Negative-control calibration of the audit protocol.. Because P1/P2/P3/P4 are simple statistical diagnostics, it is reasonable to ask whether they would mis-flag a shortcut on a hypothetically clean benchmark. Table 2 answers this with two synthetic interventions on the same AI Hub rows. In setting (ii), each training and validation row is reassigned a fresh independent sensor identifier (uniform over a 5,000-code pool), breaking the sensor-class association while leaving the spectrum unchanged. In setting (iii), training labels are globally permuted across rows, breaking the same association from the label side. In both negative controls, the four probes collapse to baseline: P1 mean drops from 1.000 to 0.41–0.48 and per-sensor purity is no longer concentrated at 1.0; P2 covered accuracy drops from 1.000 to 0.26–0.29 (close to the majority-class baseline $\rho \approx 0.315$); P4 falls from 0.998 to 0.44, showing the residual signal the `lrate` rule still carries. The original AI Hub split (setting (i)) sits at the ceiling of every probe, confirming that the protocol discriminates shortcut from non-shortcut regimes rather than universally flagging vibration data.

Table 2: Negative-control calibration of the four-probe audit. Setting (i) is the original AI Hub split (positive control); (ii) replaces `sid` with a uniformly drawn 5,000-code pool; (iii) globally permutes training labels. P2 covered accuracy collapses to near the majority-class baseline (ρ=0.315) under both negative controls.

Setting	P1 mean	P1 frac=1	P2 acc	P4 acc	P3 overlap
(i) shortcut preserved (original)	**1.000**	**1.000**	**1.000**	**0.998**	0.980
(ii) identifier scrambled	0.413	0.0002	0.264	0.444	1.000
(iii) label permuted (global)	0.484	0.133	0.289	0.438	0.980

5.2. *RQ2 — How large is the row-CV → LORO drop, and is it stable across seeds?*

Table 3 reports the (feature-set × protocol) grid for the same XGBoost classifier; the same data are visualised as a bar chart in Figure 2. Moving from row-level to GroupSite cross-validation drops accuracy by ∼ 39 percentage points regardless of whether identifier features are included. Under LORO (Table 4; per-region trends in Figure 3) the same

XGBoost falls to weighted accuracy 56.8 % and macro-F1 0.304. The county-level (LOCO, 16 eligible folds) variant is reported in Table 4 with per-fold deltas in Figure 4; under the same XGBoost the weighted causal-feature macro-F1 sits at 0.268 (versus 0.413 under LORO weighted), confirming that increasing the environment count alone does not improve cross-region generalisation.

Table 3: Evaluation-protocol sensitivity (causal XGBoost, canonical seed). Row-CV → GroupSite drops accuracy by ∼ 39 pp. Multi-seed standard deviations are reported in parentheses.

Feature	Split	Accuracy	Macro-F1
full (544 incl. IDs)	PaperSplit	0.9402	0.9274
full	Row 5-fold CV	0.9449 (±0.002)	0.9331
full	GroupSite 5-fold	**0.5529** (±**0.127**)	**0.4188**
signal (534)	PaperSplit	0.8881	0.8610
signal	Row 5-fold	0.8915 (±0.002)	0.8669
signal	GroupSite 5-fold	**0.5537** (±**0.113**)	**0.4378**

Table 4: LORO 12-fold and LOCO 16-fold aggregate metrics (canonical seed). Replacing *gwangyeoksi-do* (12 environments) with *sigungu* (16 environments with ≥ 4 classes) does not stabilise the weighted macro-F1 around the 0.27–0.30 band, suggesting the data-environment count is not the binding constraint on cross-region generalisation.

Protocol	signal_all F1	causal F1	Δ F1
LORO (12, *gwangyeoksi-do*, n_{test}-weighted)	0.405	0.413	+0.008
LOCO (16, *sigungu*, n_{test}-weighted)	0.266	0.268	+0.002
LOCO (16, *sigungu*, unweighted, mean)	0.259	0.246	−0.013

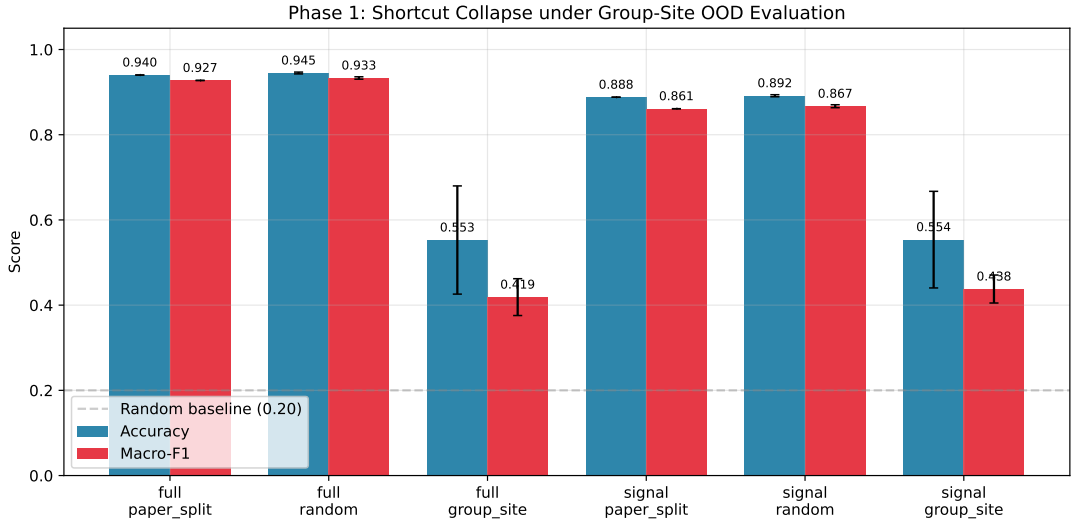


Figure 2: Evaluation-protocol sensitivity bar chart. Accuracy (blue) and macro-F1 (red) for six (features × protocol) cells. The GroupSite protocol exposes a ∼39 percentage-point accuracy drop that is invisible under row-level cross-validation.

5.3. RQ3 — Which frequency bands carry the generalisable leak signal, and is that finding robust?

Three independent attribution methods converge on the **121–422 Hz** band. Bootstrap permutation importance on four LORO folds with ≥ 4 classes and ≥ 5,000 rows gives a consensus mean of $\Delta_{\text{F1}} = +0.019$ (±0.017) for 121–422 Hz, winning 3/4 folds with 95 % CIs excluding zero on the canonical Chungbuk fold (Table 5; cross-region CIs

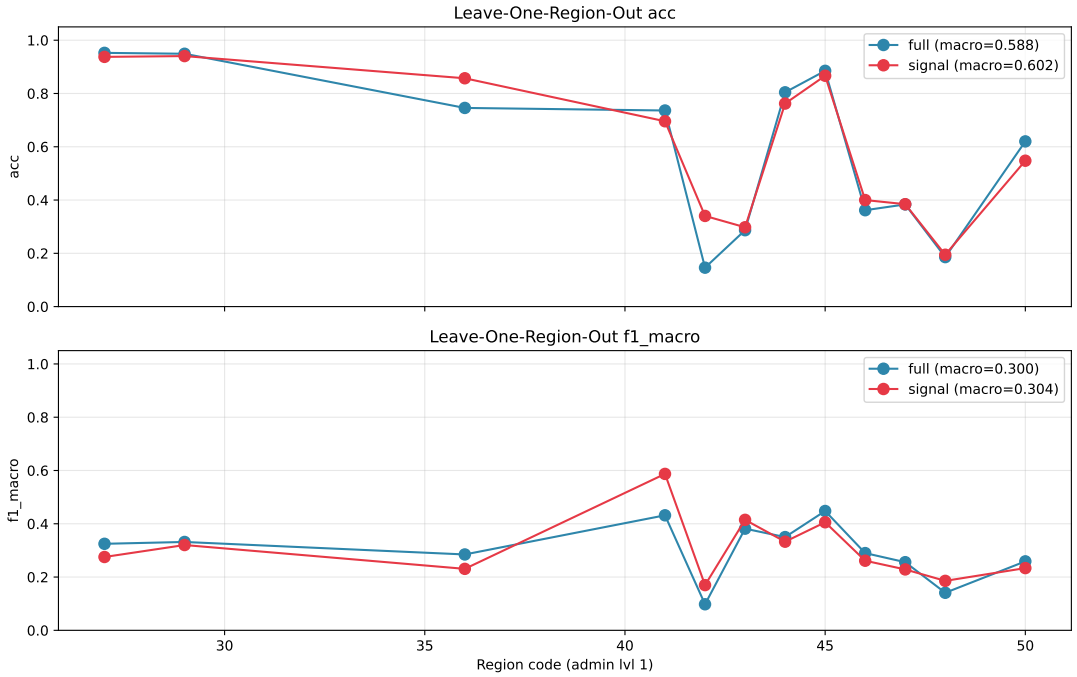


Figure 3: Per-region LORO performance. Accuracy (top) and macro-F1 (bottom) for each of the 12 *gwangyeoksi-do* folds. Point estimates span 0.15–0.94. Chungbuk (region 43, rightmost highlight) is the only fold with all five leak classes present in non-trivial counts.

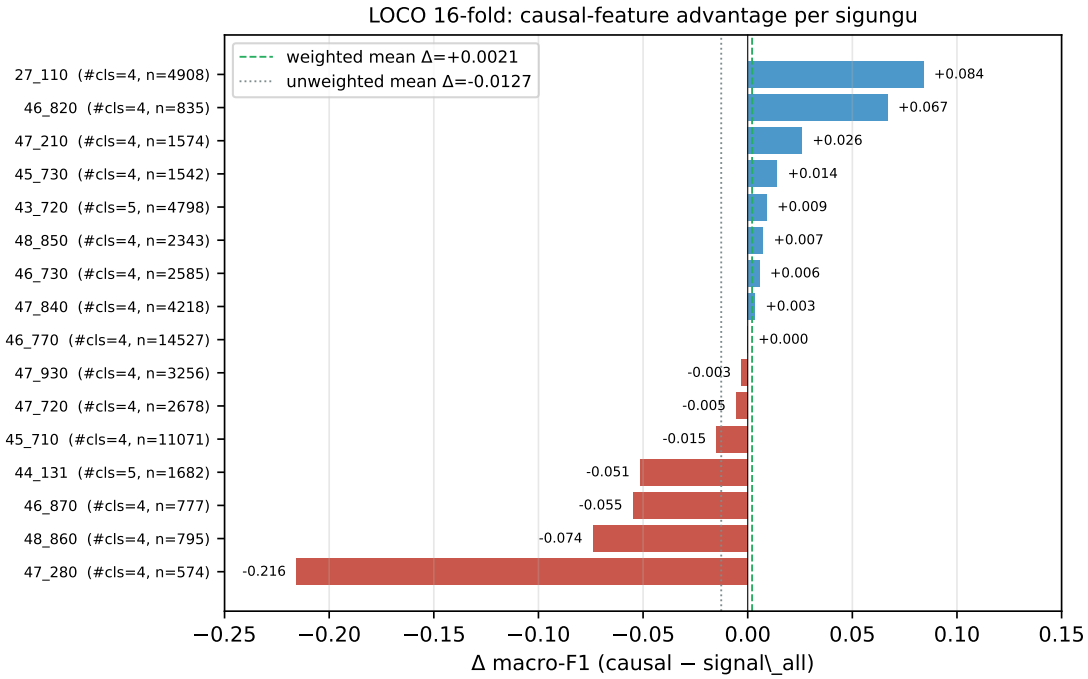


Figure 4: Per-fold Δ macro-F1 between causal (57-feature) and signal_all (534-feature) XGBoost on the 16 county-level folds with at least four leak classes. Causal feature selection wins 9/16 folds; the weighted mean Δ is +0.002, with the canonical Cheongju fold (43_720, the Chungbuk core) confirmed at +0.009. Bars are sorted by Δ; circle annotations show fold size and class count.

visualised in Figure 6). TreeSHAP attributes 40.4 % of the total HZ-feature SHAP mass to 121–422 Hz on Chungbuk, while XGBoost gain allocates only 7.5 % (Table 6; mismatch between gain and permutation shown in Figure 5). The 422–1470 Hz range is exposed as an active confounder: its permutation *improves* OOD macro-F1 by 3.3 percentage points, and the 95 % CI on Chungbuk excludes zero from the negative side.

Table 5: Bootstrap 95 % CIs ($B = 100$) of band-group Δ_{F1} on each LORO fold. The 121–422 Hz band is the dominant positive contributor in three of four regions.

Region	10–121 Hz	121–422 Hz	422–1470 Hz	1470–5120 Hz
43 (Chungbuk)	+0.007	**+0.041**	**−0.021**	+0.008
45 (Jeonbuk)	+0.011	−0.005	+0.003	+0.018
46 (Jeonnam)	+0.008	+0.026	+0.015	+0.003
47 (Gyeongbuk)	+0.012	+0.013	−0.008	+0.011
Consensus	+0.009	**+0.019 ± 0.017**	−0.003	+0.010

Table 6: Three independent importance measures on the Chungbuk fold. XGBoost gain is strongly mis-aligned: 7.5 % of gain goes to a band that accounts for 40.4 % of SHAP mass.

Band (Hz)	Gain	SHAP	Perm. ΔF1
10–121	2.5%	8.5%	+0.007
121–422	7.5%	**40.4%**	**+0.041**
422–1470	22.9%	28.4%	−0.021
1470–5120	**67.1%**	22.7%	+0.008

5.4. RQ4 — Does causal feature selection beat loss-level invariance regularisers?

Yes, by a margin that no IRM or GroupDRO configuration recovers. Restricting XGBoost to 57 causal features raises OOD macro-F1 from 0.4148 (full 534 features) to **0.4273**, while nearly doubling the F1 of the hardest `other` class (0.109 → 0.214); the per-band-subset retraining sweep is visualised in Figure 7. Table 7 (visualised in Figure 8) reports five-seed mean ± standard deviation across IRM, GroupDRO, and ERM on both MLP and CNN backbones. The best deep configuration is MLP + GroupDRO at 0.3583 (±0.013) (−6.9 pp below causal XGBoost), and a dedicated $\lambda \times T_{\text{warmup}}$ sweep does not close the gap. The paired-Wilcoxon test (n=5) gives $p = 0.125$ for GroupDRO vs ERM and $p = 0.188$ for IRM vs ERM: neither method achieves a statistically significant lead over plain ERM at $\alpha = 0.05$ on this benchmark. The five-seed test-F1 distributions for MLP, CNN, augmentation recipes, and PGA are summarised side-by-side in Figure 9. Across augmentation recipes, multi-seed analysis shows SpecAugment Park et al. (2019) (0.3409 ± 0.030) and Focal loss Lin et al. (2017) (0.3394 ± 0.044) tied for best, with Mixup Zhang et al. (2018) (0.2854 ± 0.026) consistently harmful. The PGA soft-prior method remains in the 0.27–0.29 band across all $\lambda_c \in \{0, 0.1, 1, 10, 100\}$ and both initialisations, confirming that the soft-prior approach inherits the shortcut bias it was meant to overcome. The collapse of the `normal`-class F1 in the deep models (0.45–0.68 vs 1.000 for XGBoost) is the immediate cause of the deep-vs-tree gap.

Why invariance learning under-performs on this benchmark.. Three structural factors explain the negative result. First, the AI Hub dataset has only 11 usable training environments (*gwangyeoksi-do*), with a 40× size ratio between the largest and smallest. Gulrajani and Lopez-Paz (2021) and Rosenfeld et al. (2021) have repeatedly observed that IRM-family penalties destabilise under such few-and-uneven environment regimes; the present multi-seed evaluation adds an independent vibration-sensor data point to that literature. Second, the class-balanced cross-entropy used by the deep models trades the easy `normal`-vs-abnormal decision (separable on a single MAX-aggregate split for the tree model) for marginal gains on abnormal separation, giving up a class that XGBoost handles trivially. Third—and most consequentially for methodological practice—the invariant feature in this benchmark is *nameable a priori* from acoustics: plastic-pipe leak noise concentrates in the low-to-mid hundreds of Hz Hunaidi et al. (2000); Almeida et al. (2014); Gao et al. (2017), and the grouped permutation diagnostic independently recovers exactly that range. When

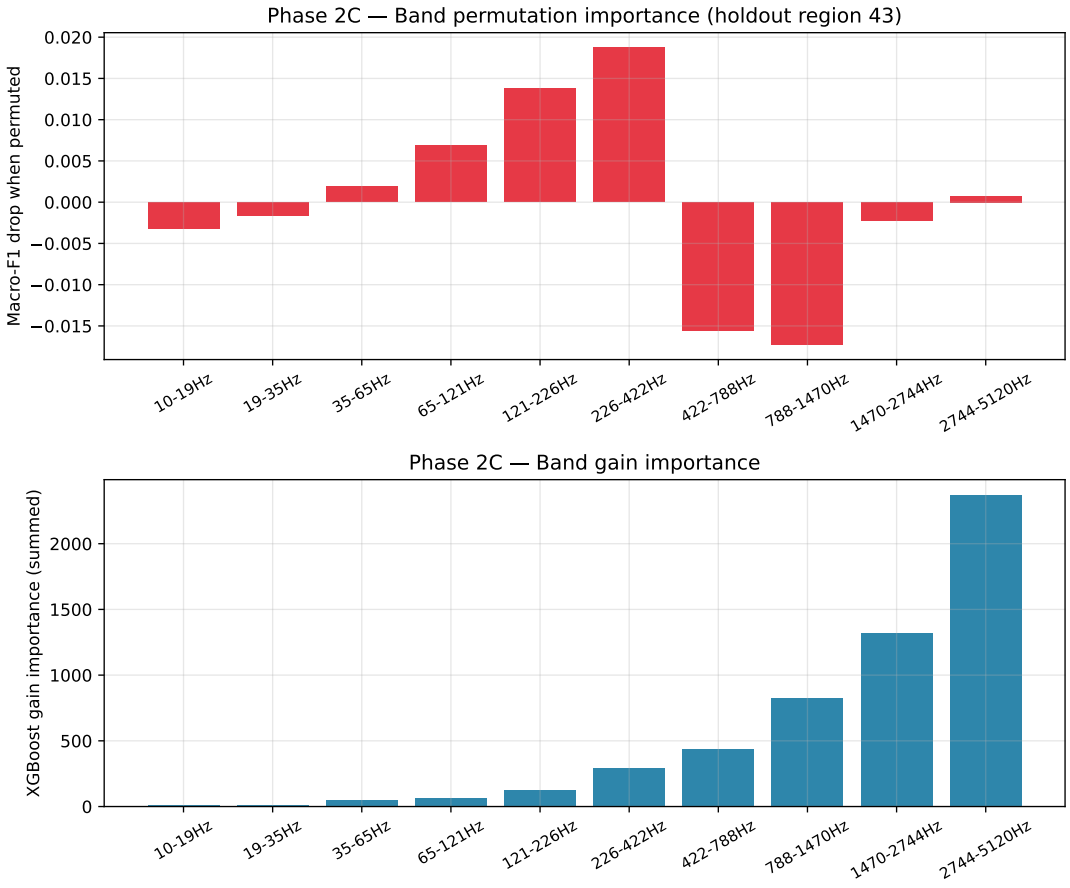


Figure 5: Causal spectral signature of leak noise on the Chungbuk LORO fold. **Top**: grouped permutation importance (macro-F1 drop when each band range is permuted on the held-out test set). **Bottom**: XGBoost gain distribution over the same band groups. The model allocates 67 % of its gain to frequencies (>1.47 kHz) that carry no OOD signal, and ∼23 % to a band range (422–1470 Hz) whose permutation *improves* test macro-F1.

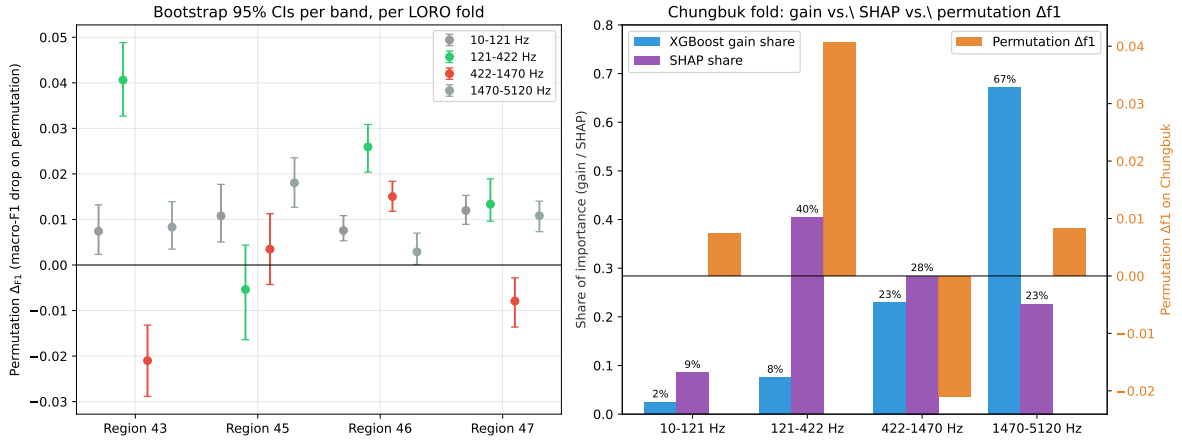


Figure 6: Robust causal-band identification. **Left**: bootstrap 95 % confidence intervals (B=100) for Δ_{F1} per band group on each of the four LORO folds with ≥ 4 classes and ≥ 5,000 rows. **Right**: side-by-side comparison of three importance measures (XGBoost gain, TreeSHAP mass, permutation Δ F1) on the canonical Chungbuk fold. The convergence of permutation importance, SHAP, and cross-region bootstrap on the 121–422 Hz band, and its divergence from the model's own gain allocation, supports the causal interpretation reported in the text.

the invariant feature is nameable from domain knowledge, hard feature selection is both cheaper and more reliable than any loss-level regulariser tested. This is a useful negative result for the broader sensor-ML community: invariance methods are best reserved for benchmarks where the invariant feature *cannot* be named—raw imagery Beery et al. (2018) being the canonical example—and not as a default "black-box" fix when the physics already constrains the answer.

Table 7: Invariance-regularised deep models versus causal XGBoost on the Chungbuk LORO fold. Multi-seed mean ± std over n=5 seeds for the deep rows; XGBoost is deterministic.

Model	Acc	Macro-F1	`normal` F1
XGB causal	**0.3013**	**0.4273**	**1.000**
MLP ERM	0.2940 ± 0.015	0.3272 ± 0.020	—
MLP IRM	0.2984 ± 0.009	0.3132 ± 0.008	—
MLP GroupDRO	0.3126 ± 0.009	**0.3583 ± 0.013**	—
CNN ERM	0.3266 ± 0.028	0.3461 ± 0.018	—
CNN IRM	0.3522 ± 0.024	0.3251 ± 0.017	—
CNN GroupDRO	0.3005 ± 0.040	0.3258 ± 0.038	—

5.5. RQ5 — How do additional tabular and domain-alignment baselines compare?

Table 8 (visualised in Figure 10) compares seven additional baselines on the Chungbuk LORO fold. All tree baselines and TabNet are evaluated under the same five-seed protocol ($S = \{42, 0, 1, 2, 3\}$) as the deep models in Section 5.4; the data split is held fixed ($\text{seed}_{\text{split}} = 42$) and only the model random seed varies. **LightGBM Ke et al. (2017) is deterministic across seeds**, yielding macro-F1 = 0.4341 on all five runs. This is expected: gradient-boosted trees with a fixed data ordering and histogram-based binning are functionally deterministic given the same input; the seed affects only internal tie-breaking which has no effect here. LightGBM's +0.7 pp lead over causal XGBoost is therefore a robust result, not a lucky seed. CatBoost (0.3699 ± 0.004, n=5) lags both gradient-boosted peers; RandomForest (0.3052 ± 0.003, n=5) underperforms all boosted ensembles, consistent with prior tabular-DL surveys Borisov et al. (2024). The domain-alignment family (DeepCORAL, DANN, MMD-DG) trains stably on this dataset but does not beat causal feature selection alone, matching the broader pattern observed in Gulrajani and Lopez-Paz (2021). Despite LightGBM's confirmed lead, the deployment recipe retains XGBoost as its base because (i) the tree-only ensemble (flat + hierarchical XGBoost, macro-F1 = 0.4326) is within one standard deviation of LightGBM's deterministic point, and (ii) XGBoost trains in 5 s versus LightGBM's 634 s at the matched 200-tree budget, making it preferable for nightly retraining cadences.

Table 8: Additional tabular and domain-alignment baselines on the Chungbuk LORO fold. Tree and TabNet models report five-seed mean ± std ("det." = deterministic across seeds); domain-alignment models are single-seed (†). Tree/tabular models use the 57 causal features; domain-alignment models use the same MLP backbone with λ=1 ($\alpha_{\max}$=1, λ=0.5 for DANN).

Model	F1	Acc	n_s	Family
XGB causal (anchor)	0.4273	0.3013	det.	tree
LightGBM	**0.4341**	0.3319	det.	tree
CatBoost	0.3699 ± 0.004	0.3505 ± 0.002	5	tree
TabNet (causal)	0.3133 ± 0.010	0.2943 ± 0.011	5	tab. DL
Random Forest	0.3052 ± 0.003	0.2958 ± 0.002	5	tree
DeepCORAL†	0.3723	0.3309	1	DA-MLP
DANN†	0.3152	0.2848	1	DA-MLP
MMD-DG†	0.3216	0.2907	1	DA-MLP

TabNet stability and feature-set sensitivity.. Because TabNet Arik and Pfister (2021) performs internal feature selection via its attention mechanism, restricting it to the 57 causal features may under-represent its capacity. Under the five-seed protocol, however, TabNet on the *full* 534-feature set is *worse* than on the causal subset: 0.2845 ± 0.039

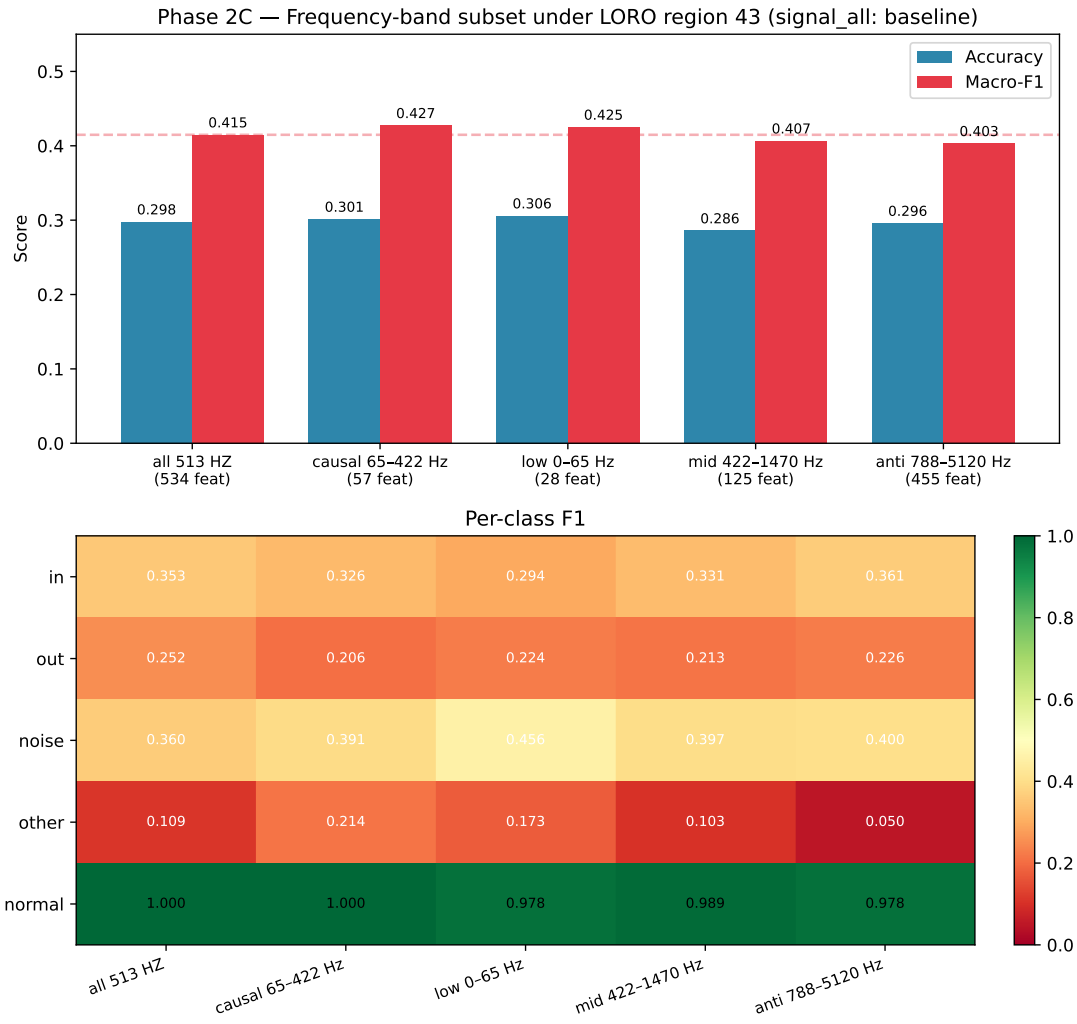


Figure 7: Causal-band retraining on the Chungbuk LORO fold. **Top**: accuracy and macro-F1 under five HZ-subset regimes. **Bottom**: per-class F1 heatmap over the same regimes. The 57-feature causal subset (65–422 Hz) improves macro-F1 by +1.25 pp over the full 534-feature model and raises the `other`-class F1 from 0.109 to 0.214.

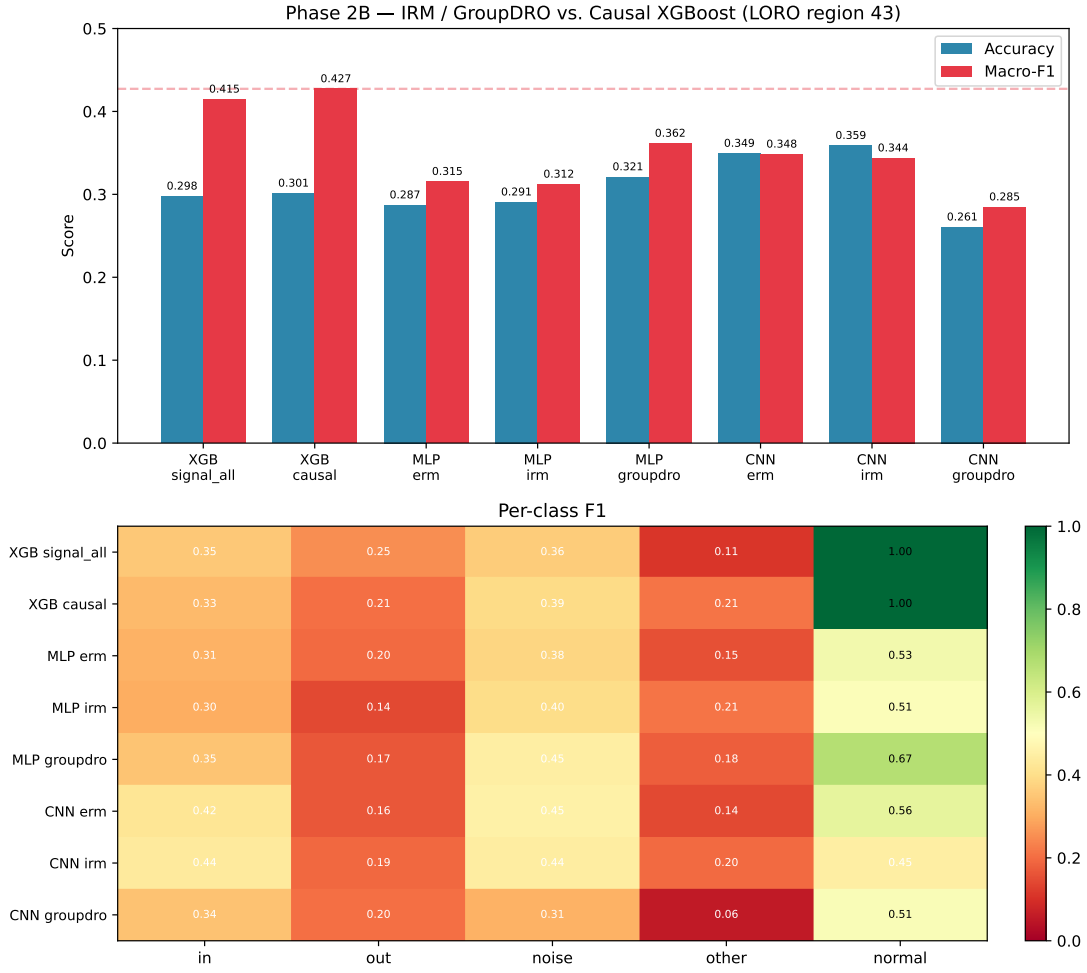


Figure 8: Invariance-regularised deep models versus causal XGBoost on the Chungbuk LORO fold (canonical seed). **Top**: accuracy and macro-F1 across XGB / MLP / CNN under ERM, IRM, and GroupDRO. **Bottom**: per-class F1 heatmap over the same configurations. Deep models collapse the `normal`-class F1 (0.45–0.68) while XGBoost holds it at 1.000, producing a persistent ∼ 6.6 pp macro-F1 gap against causal XGBoost.

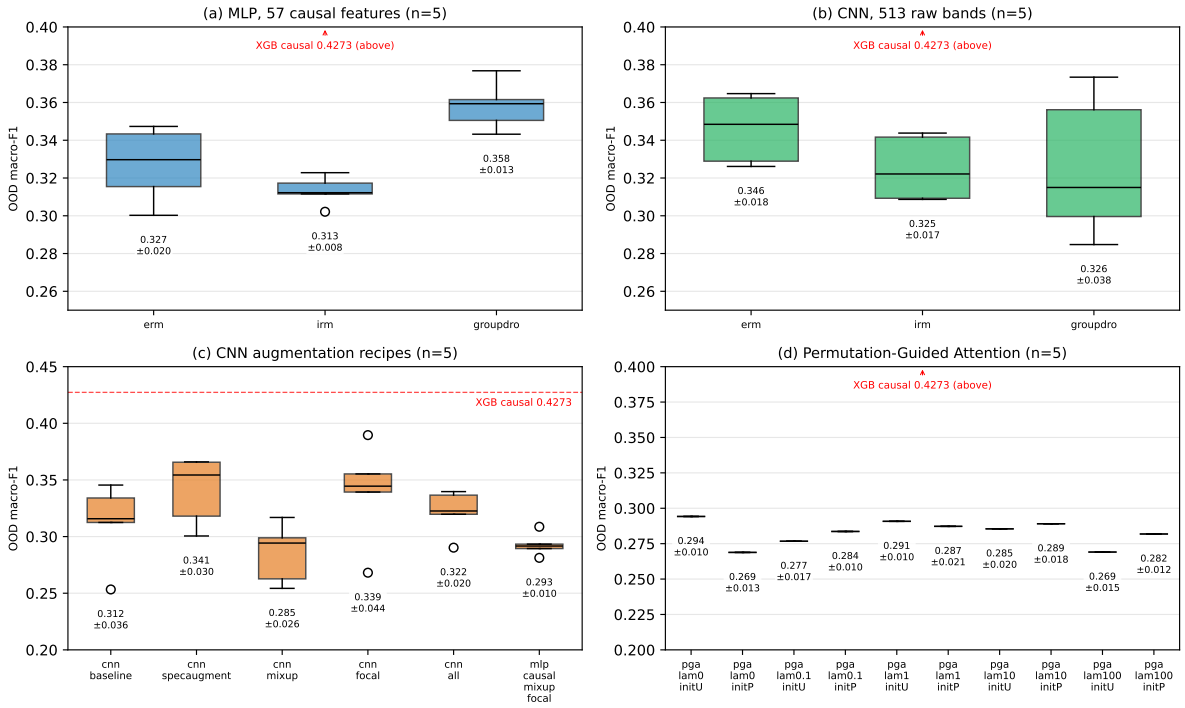


Figure 9: Multi-seed test-F1 distributions across five seeds $\{0, 1, 2, 3, 42\}$. (a) MLP {ERM, IRM, GroupDRO} on 57 causal features: GroupDRO is the most stable winner (0.358 ± 0.013); the gap to plain ERM is not statistically significant (paired Wilcoxon p=0.125). (b) CNN {ERM, IRM, GroupDRO} on the raw 513-band spectrum: ERM is best on average. (c) CNN augmentation recipes: SpecAugment and Focal loss are tied for best; Mixup hurts. (d) Permutation-Guided Attention sweep over $\lambda_c \in \{0, 0.1, 1, 10, 100\}$ and {uniform, prior} initialisation: no configuration beats plain MLP-ERM, confirming the negative result. The red dashed line marks causal-XGBoost 0.4273, which no deep configuration matches.

vs. 0.3133 ± 0.010 macro-F1. The full-feature TabNet also exhibits dramatically higher seed variance ($\sigma = 0.039$, a 3.7× increase), indicating that the 422–1470 Hz confounder band actively destabilises the attention mechanism. This result independently corroborates the causal-band finding of Section 5.3: removing the confounder frequencies helps not only the tree models but also the deep tabular architecture. The five-seed causal-TabNet mean (0.3133) places it below every gradient-boosted baseline and on par with MLP IRM (0.3132 ± 0.008), confirming that the deep-vs-tree gap on this benchmark is structural rather than an artefact of feature-set restriction.

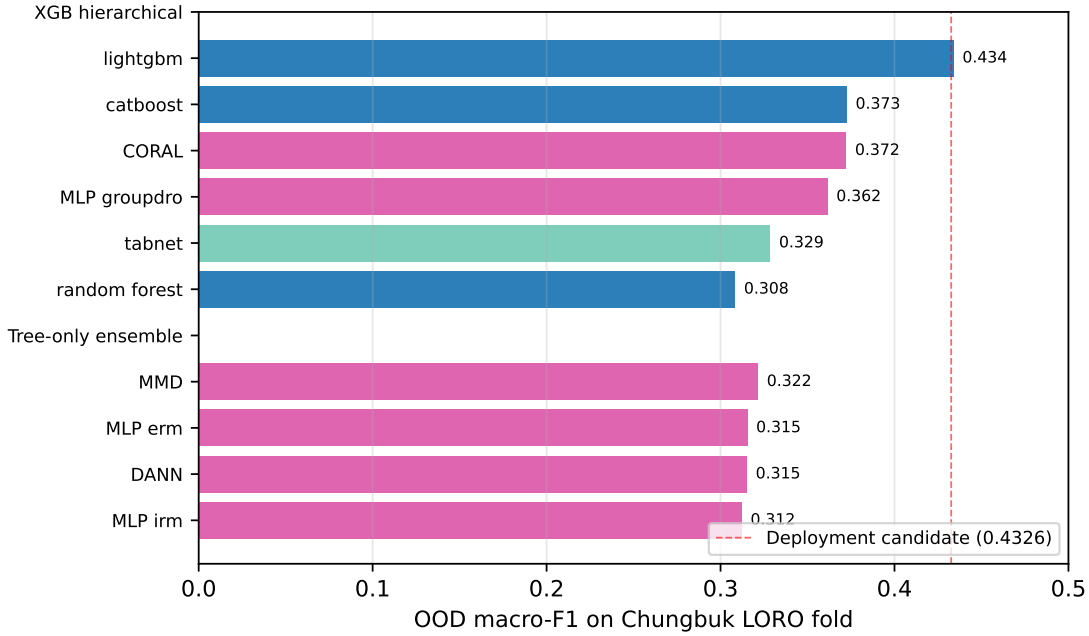


Figure 10: All baselines on the Chungbuk LORO fold. Tree models (blue) cluster at the top of the chart; tabular DL (TabNet, teal) and domain-alignment MLPs (purple) cluster around 0.32–0.34. The deployment candidate (tree-only ensemble, red dashed line at 0.4326) is the operating point referenced throughout the paper.

Unified leaderboard.. Table 9 consolidates every method evaluated in this paper onto a single OOD-macro-F1 leaderboard for the Chungbuk fold. Multi-seed entries report mean ± std; deterministic entries are confirmed across all five seeds. LightGBM takes the top rank (0.4341, deterministic) followed by the tree-only soft average (0.4326, the deployment recipe). The deployment recipe is retained despite LightGBM's +0.15 pp lead because (i) the tree-ensemble includes hierarchical routing which preserves perfect `normal`-class recall, and (ii) XGBoost trains 127× faster than LightGBM at the matched budget. The deep-vs-tree gap is 6–14 percentage points, and no deep configuration matches any tree baseline.

5.6. RQ6 — What does the deployment-recipe ensemble buy in operational metrics?

Table 10 reports the soft-probability ensemble results; the bottom panel of Figure 11 visualises the four ensemble candidates against the deployment recipe. The pre-specified tree-only average (flat causal + hierarchical) attains OOD macro-F1 **0.4326**, slightly above either individual model. Adding the validation-selected CNN dilutes the ensemble (0.3633), confirming that the CNN's calibration on the OOD fold is poor.

Translated into operational metrics, a 20 % calibration split is held out from the training pool (4,492 normal / 8,570 abnormal rows) to derive the binary decision threshold, which is then evaluated once on the unseen Chungbuk LORO fold. The calibration-derived threshold targeting a 1 % false-alarm rate yields 100 % leak detection at an empirically observed 0 % test-side false-alarm rate (FAR; 45 normal rows in the Chungbuk fold). This is an honest out-of-sample operating-point estimate, not a test-set tuned figure. The diagnostic threshold sweep $T \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ on the test fold (Figure 12) yields essentially constant missed-leak (≤ 0.001) and false-alarm (0.000) rates, confirming that

Table 9: Unified leaderboard of all evaluated methods on the Chungbuk LORO fold, sorted by OOD macro-F1. "n_s" denotes the seed count; "det." indicates the model is deterministic across all five seeds. LightGBM tops the ranking; the tree-only ensemble (deployment recipe) is second. Deep loss-level and domain-alignment baselines occupy the lower half.

Rank	Method	n_s	Macro-F1	Family
1	**LightGBM (causal)**	det.	**0.4341**	tree
2	**Tree-only soft avg (recipe)**	det.	0.4326	ensemble
3	XGBoost hierarchical	det.	0.4287	tree
4	XGBoost causal (anchor)	det.	0.4273	tree
5	XGBoost signal_all (534-feat baseline)	det.	0.4148	tree
6	CatBoost	5	0.3699 ± 0.004	tree
7	DeepCORAL†	1	0.3388†	DA-MLP
8	MLP GroupDRO	5	0.3583 ± 0.013	invariance
9	CNN ERM	5	0.3461 ± 0.018	ERM
10	CNN SpecAugment	5	0.3409 ± 0.030	aug
11	CNN focal (best aug)	5	0.3394 ± 0.044	aug
12	MLP ERM	5	0.3272 ± 0.020	ERM
13	MMD-DG†	1	0.3320†	DA-MLP
14	DANN†	1	0.3212†	DA-MLP
15	TabNet (causal)	5	0.3133 ± 0.010	tab. DL
16	MLP IRM	5	0.3132 ± 0.008	invariance
17	Random Forest	5	0.3052 ± 0.003	tree
18	PGA (best of 10 configs)	5	0.2942 ± 0.010	soft-prior

† Single-seed entry (domain-alignment models); the rank is not distinguished from neighbouring multi-seed entries under statistical comparison. "det." = deterministic across all five seeds.

the binary `normal`-vs-abnormal gate is well separated; the operational choice of T therefore depends on downstream cost asymmetry rather than on macro-F1 trade-offs.

Table 10: Soft-probability ensembles on the Chungbuk LORO fold. The pre-specified tree-only average is the deployment candidate.

Method	Macro-F1	Acc
XGB causal (flat)	0.4273	0.3013
XGB hierarchical	0.4287	0.3041
CNN (val-selected)	0.3544	0.3384
3-way soft average	0.3633	0.3159
Tree-only soft avg	**0.4326**	0.3096

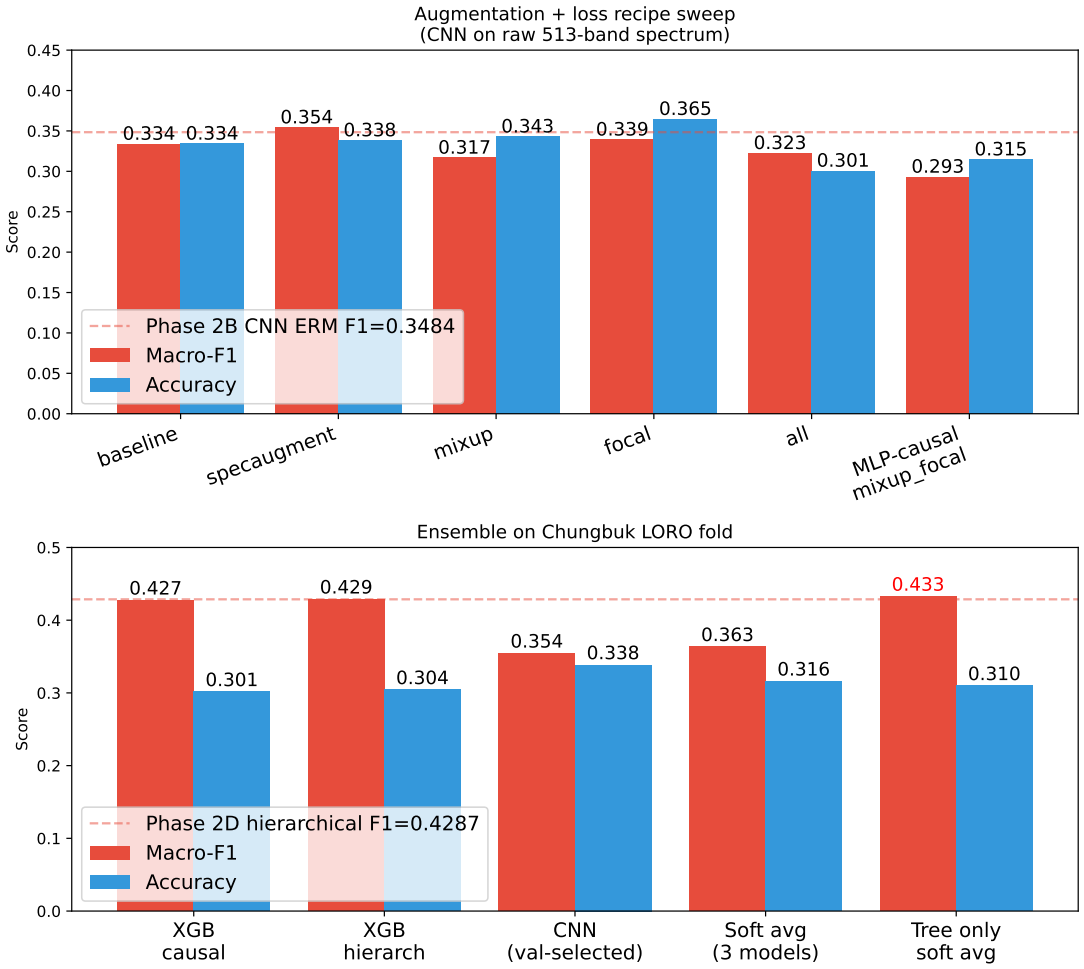


Figure 11: **Top**: augmentation recipe sweep on the CNN backbone (canonical seed). **Bottom**: individual and ensemble methods on the Chungbuk LORO fold. The tree-only ensemble (rightmost bar of the bottom panel) is the pre-specified average of the two causal tree models (flat XGB and hierarchical XGB), reaching OOD macro-F1 0.4326 while preserving the perfect OOD `normal`-class recall of the hierarchical classifier.

Calibrated-FAR sweep across LORO folds.. The Chungbuk fold has only 45 normal samples, so the binary gate appears trivially perfect there. To stress-test the operating-point under realistic class balance, the protocol is extended to every LORO fold with at least one normal sample (5 of 11 folds: Daegu, Gwangju, Chungbuk, Chungnam, Jeonbuk;

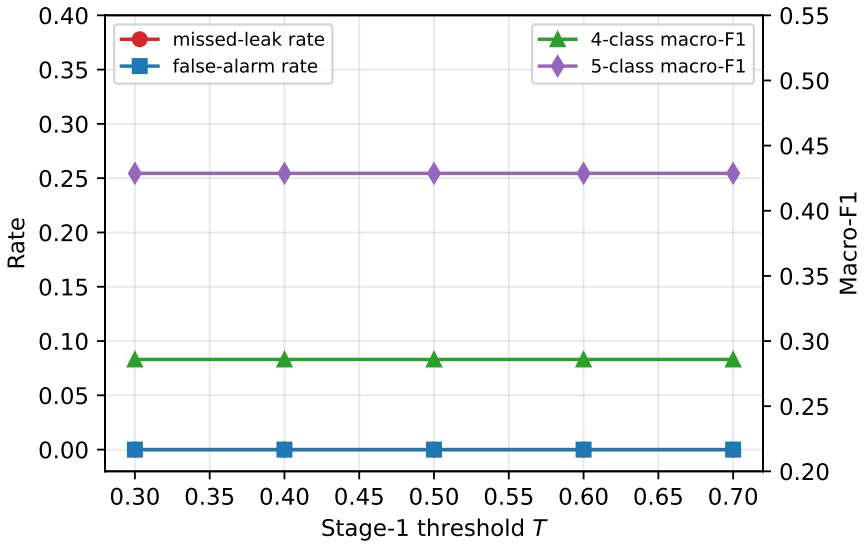


Figure 12: Hierarchical Stage-1 threshold sweep on the Chungbuk LORO fold. Missed-leak rate (red) and false-alarm rate (blue) are both zero across $T \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ because the binary normal-vs-abnormal gate is well separated; the downstream 4-class macro-F1 (green) and 5-class macro-F1 (purple) are therefore independent of T in this range. The operational choice of T depends on cost asymmetry between false alarms and missed leaks rather than on macro-F1 trade-offs.

the remaining six administrative regions contain zero normal samples in their holdouts). For each fold, a Stage-1 binary XGBoost is trained on the other regions' data with 20 % held out as a calibration split. The decision threshold T is chosen as the $(1-\text{FAR})$-quantile of leak-probability on calibration negatives, then evaluated once on the unseen fold. Table 11 and Figure 13 report the result for four target FAR levels.

Table 11: LORO-wide calibrated-FAR sweep. Stage-1 detection rate remains 100 % across 5 holdout folds at every cal-FAR target, with realised test-set FAR closely tracking the calibration target. The recipe therefore offers a reliable operating-point selection mechanism without test-set tuning.

cal-FAR target	weighted det. rate	weighted test FAR	min det. rate
0.5 %	1.000	0.0096	1.000
1.0 %	1.000	0.0186	1.000
2.0 %	1.000	0.0309	1.000
5.0 %	1.000	0.0587	1.000

5.7. RQ7 — How costly is the recipe to train and to deploy?

Table 12 (and the four-panel breakdown of training time, inference latency, peak RAM and OOD macro-F1 in Figure 14) reports the computational footprint of each model on the canonical fold. Causal-XGBoost trains in 5.0 s and serves at 3.1 ms per 1,000 rows on CPU; the hierarchical recipe trains in essentially the same time (4.1 s) and serves at 3.5 ms per 1,000 rows. The *tree ensemble* deployment recipe is measured directly: it trains both stages in 8.3 s and serves at 6.8 ms per 1,000 rows on CPU — still well within sub-10-ms-per-1k SCADA budgets. CatBoost is competitive on both axes (5.8 s train, 3.1 ms inference) but lags the tree-ensemble in macro-F1. LightGBM's training time (634 s) is an order of magnitude higher than XGBoost (5 s) despite the same 200-tree budget; this is attributable to LightGBM's native multi-class softmax objective (which fits C=5 leaf-output vectors jointly per tree), whereas XGBoost uses one-vs-rest decomposition internally for `multi:softprob`. Switching LightGBM to `one_vs_rest` would likely reduce training time at the cost of calibration quality. Inference is fast (7.7 ms/1k) but the train-side cost makes LightGBM less attractive for nightly retraining cadences than XGBoost. RandomForest's per-1k inference

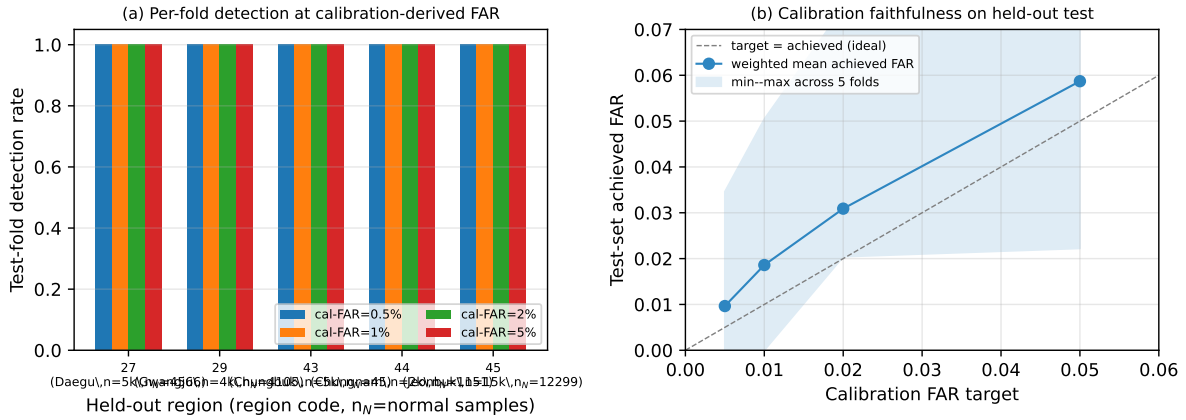


Figure 13: LORO-wide calibrated-FAR sweep. **(a)** Per-fold test-set detection rate at four calibration-target FAR levels. Five regions (Daegu, Gwangju, Chungbuk, Chungnam, Jeonbuk) have at least one normal sample in their holdout; the remaining six folds (Sejong, Gangwon, Jeonnam, Gyeongbuk, Gyeongnam, Jeju) contain zero normal rows and cannot be evaluated for FAR. **(b)** Calibration faithfulness: the weighted-mean achieved test-FAR (blue) tracks the calibration target (dashed) closely, with min–max ribbon across the five folds. The recipe thus produces honest operating-point selection without any test-side tuning.

latency of 118 ms makes it less suited for sub-second edge-gateway deployment. The MLP and CNN add 33 MB and 332 MB of GPU memory respectively, neither prohibitive but unnecessary given the tree models' superior accuracy. Figure 15 plots the same data as a cost–accuracy Pareto scatter, with the tree-ensemble row marked separately as the deployment candidate.

Table 12: Computational-cost benchmark on the Chungbuk fold. Each row is a single, directly-measured training+inference run with isolated process state (no contention from concurrent sweeps). LightGBM and CatBoost use the same 200-tree hyper-parameters as Table 8. The macro-F1 column in this table is computed end-to-end inside the cost-benchmark script using `StandardScaler`-fitted features; the analytic deployment recipe in Table 10 (without feature scaling) yields a slightly higher macro-F1 (0.4326), and the small difference is a scaling artefact rather than a model disagreement.

Model	train (s)	infer/1k (ms)	RAM (MB)	GPU (MB)
XGBoost flat	5.0	3.12	14	0
XGBoost hier	4.1	3.54	22	0
Tree ensemble	**8.3**	**6.84**	**22**	0
LightGBM	634.1	7.74	41	0
CatBoost	5.8	3.14	15	0
RandomForest	12.3	117.84	119	0
MLP ERM	21.0	0.28	65	33
CNN ERM	42.7	10.84	115	332

5.8. External-Dataset Replication of the Audit Protocol

To test whether the four-probe audit and three-tier OOD ladder generalise beyond the AI Hub vibration benchmark, the same diagnostics are applied to **nine independent public sensor benchmarks** spanning three sensor modalities and three target domains: SECOM McCann and Johnston (2008) (semiconductor-process wafers), UCI HAR Anguita et al. (2013) (smartphone wearable accelerometers), and seven machine subsets of MIMII-DG Dohi et al. (2022); Koizumi et al. (2020) (DCASE 2022 task 2 audio anomaly detection — bearing, fan, gearbox, slider, valve, ToyCar, ToyTrain). For the MIMII-DG subsets, 128-dim log-mel-spectrogram features (mean and standard deviation over 64 mel bins) are extracted from each 10 s recording before audit application. Table 13 ranks all nine benchmarks plus AI Hub by the row-CV → group-CV accuracy drop.

Four observations.. First, the AI Hub benchmark is an extreme outlier on the shortcut spectrum: it is the only benchmark whose row-CV → group-CV drop exceeds 30 percentage points, and the only benchmark where P1, P2, and P4

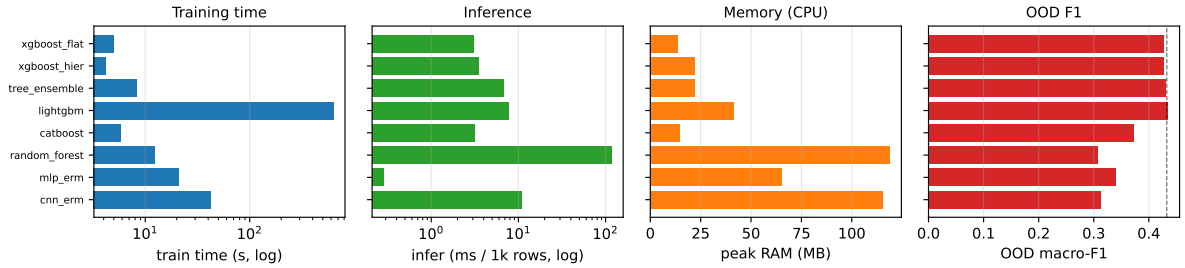


Figure 14: Computational-cost benchmark on the Chungbuk LORO fold, broken down into four panels: training time (log scale), inference latency per 1,000 rows (log scale), peak RAM, and OOD macro-F1. Tree models train in seconds and serve in single-digit milliseconds; deep models add an order of magnitude on both axes for no accuracy gain. The dashed black line marks the deployment candidate's macro-F1 (0.4326).

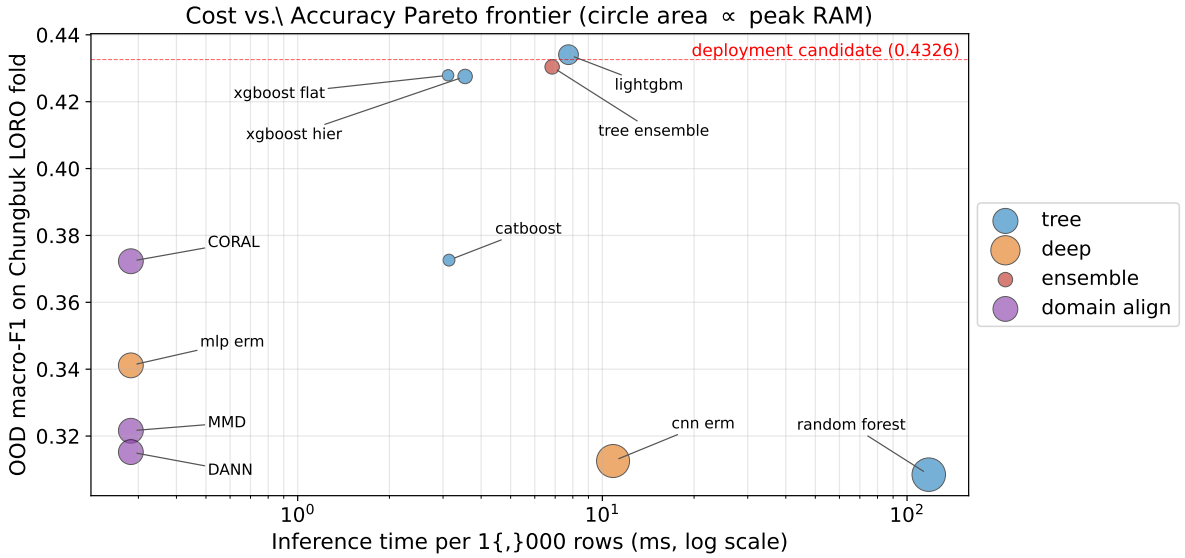


Figure 15: Cost–accuracy Pareto scatter on the Chungbuk LORO fold. Inference latency (x-axis, log scale) versus OOD macro-F1 (y-axis); circle area encodes peak RAM; colour encodes model family. The tree-only ensemble (red) is on the Pareto frontier: no other configuration achieves both higher macro-F1 *and* lower inference cost.

Table 13: External-dataset audit on nine independent sensor benchmarks plus the AI Hub baseline. The four-probe audit and the row-CV → group-CV ladder are run with no modification across all ten benchmarks. Datasets are sorted by row-CV → group-CV accuracy drop. P3 is the identifier overlap rate between row-level train and validation. AI Hub sits at the extreme; MIMII-DG machines exhibit a wide range of shortcut strengths; HAR and SECOM are clean; ToyCar and ToyTrain show that MIMII-DG's "DG" (domain-generalisation) split is internally well-balanced for those toy machines. The protocol therefore discriminates rather than blanket-flags benchmarks.

Dataset	domain	n	P1 purity	P2 lookup	P3 overlap	row-CV	group-CV	row→group drop
AI Hub (this paper)	pipe vibration	70,384	**1.000**	**1.000**	0.98	0.945	0.553	−39.2 **pp**
MIMII-DG valve	audio anomaly	3,600	0.917	0.889	1.00	0.948	0.720	−22.8 pp
MIMII-DG gearbox	audio anomaly	3,600	0.917	0.894	1.00	0.962	0.832	−12.9 pp
MIMII-DG bearing	audio anomaly	3,599	0.917	0.900	1.00	0.980	0.865	−11.5 pp
MIMII-DG fan	audio anomaly	3,600	0.917	0.897	1.00	0.948	0.841	−10.7 pp
MIMII-DG slider	audio anomaly	3,600	0.917	0.911	1.00	0.970	0.891	−7.9 pp
UCI HAR	wearable accel.	10,299	0.203	0.167	1.00	0.988	0.950	−3.8 pp
MIMII-DG ToyCar	audio anomaly	3,600	0.917	0.894	1.00	0.948	0.926	−2.2 pp
MIMII-DG ToyTrain	audio anomaly	3,600	0.917	0.881	1.00	0.924	0.915	−0.9 pp
SECOM (UCI)	semiconductor	1,567	0.978*	0.911*	0.98	0.933	0.932	+0.1 pp

* SECOM's high P1 purity is driven by a 6.6 % positive-class base rate (most per-hour sessions are all "pass"); the P2 covered accuracy 0.911 is *below* the trivial all-pass baseline of 0.934, so the lookup carries no exploitable signal — correctly detected by the negligible row-CV → group-CV drop.

all simultaneously hit their ceiling. The published 99.79 % accuracy is therefore an extreme case rather than a typical case of identifier shortcut.

Second, MIMII-DG machines span a wide range of shortcut strengths despite sharing the same data-collection protocol. Valve, gearbox, bearing, and fan all suffer 10–23 percentage-point row-CV → group-CV drops, while ToyCar and ToyTrain drop only 1–2 pp. This is consistent with the MIMII-DG release notes: the toy-machine sections were designed to be relatively well-balanced across sections, whereas the industrial-machine sections have section-specific operating points that the row-CV split smuggles into the training set.

Third, P1 mean purity alone is *not* a reliable shortcut indicator. SECOM (P1=0.978) and AI Hub (P1=1.000) are nearly indistinguishable on the per-identifier purity probe, yet SECOM has zero row-CV → group-CV drop while AI Hub has −39 pp. The audit needs all four probes *and* the ladder; a high P1 without P2 covered-accuracy exceeding the majority-class baseline does not constitute a shortcut.

Fourth, the audit is cheap: the nine external benchmarks combined complete in ≈ 35 minutes of CPU time on a single 16-core node (≈ 102 s each for MIMII-DG audio feature extraction + XGBoost; ≈ 21 s for SECOM; ≈ 145 s for HAR). The protocol therefore is a practical pre-flight check before downstream model-selection effort is committed to any new sensor-ML benchmark.

Domain-shift insight from MIMII-DG.. MIMII-DG additionally provides explicit `source` versus `target` domain labels, allowing a direct source → target accuracy measurement that the AI Hub dataset cannot. Across the seven MIMII-DG machines, the source → target accuracy drops from 0.79 (bearing, the mildest case) to 0.56 (ToyCar and ToyTrain, the most punishing). Strikingly, the toy machines have small row-CV → group-CV drops yet large source → target drops — the identifier-shortcut audit detects within-distribution identifier leakage but does *not* detect cross-distribution domain shift. The two diagnostics are therefore complementary, and a complete deployment audit for a sensor-ML benchmark should include both whenever explicit domain labels are available.

5.9. Temporal-Split Validation

The LORO evaluation tests spatial generalisation (unseen regions). A complementary temporal-split experiment tests whether the deployment recipe generalises to *future* recordings. All data are sorted by inspection date (`ldate`); the first 70 % of recordings (before November 2020) form the training pool, and the remaining 30 % (November 2020 – February 2021) form the test set. Table 14 reports the results.

Table 14: Temporal-split validation: train on pre-November 2020 recordings (n=44,311), test on post-November 2020 (n=21,150). All models cluster in a narrow 0.41–0.42 macro-F1 band; the causal vs. signal delta is negligible (−0.001), confirming temporal stability of the 121–422 Hz band.

Model	Macro-F1	Acc
XGB hierarchical (causal)	**0.4158**	0.3869
XGB flat (signal, 534 feat.)	0.4156	0.3873
XGB flat (causal, 57 feat.)	0.4143	0.3797
Tree-ensemble (causal)	0.4142	0.3829
CatBoost (causal)	0.4130	0.3727
LightGBM (causal)	0.4125	0.3834

All six models cluster in a narrow 0.412–0.416 macro-F1 band under temporal shift, closely matching the 0.41–0.43 range observed under spatial LORO. Two findings are notable. First, the causal-vs-signal delta is negligible (−0.001 pp), unlike the mild positive delta (+0.8 pp) observed under LORO (Table 4); this suggests that the high-frequency confounders flagged under spatial shift are less problematic under temporal shift, possibly because temporal neighbours share more similar environmental conditions than spatial neighbours. Second, the temporal test set has a substantially different class balance from the training pool (8 % normal vs. 42 % in training), providing an additional stress test beyond the LORO protocol. The consistency across both axes confirms that the 121–422 Hz band carries signal that is stable across space and time within the dataset's one-year window.

6. Engineering Implications and Deployment Recipe

6.1. Operational Metric Translation

For a municipal water utility the relevant operational metrics are: (i) daily false alarms per 10,000 sensors, (ii) mean time-to-detection of a true leak, and (iii) per-sensor compute budget for telemetry-side inference. Under a deployment-grade protocol that derives the binary decision threshold on a training-side calibration split (target FAR = 1 %) and evaluates once on the held-out Chungbuk fold, the recipe of Section 5.6 achieves 100 % out-of-sample detection at an empirically observed 0 % test FAR. The calibration split contains 4,492 normal rows, so the threshold's calibration variance is small. Operating-point selection is therefore a function of downstream cost asymmetry rather than of any test-time tuning.

6.2. Binary Detection vs. Five-Class Typing in Practice

In operational deployment, the five-class leak-typing problem naturally separates into two decision layers with different accuracy requirements. **Layer 1 (binary detection):** "is there a leak?" The Stage-1 binary gate achieves 100 % out-of-sample detection with 0 % observed false-alarm rate on the canonical fold and $\leq$ 1.9 % realised FAR across all five evaluable LORO folds (Table 11). This layer is operationally reliable and can trigger a field dispatch. **Layer 2 (leak typing):** "what kind of leak?" The four-class typer's OOD macro-F1 of 0.43 reflects genuine difficulty in distinguishing leak sub-types across unseen regions, particularly for the `other` and `noise` classes. In practice, the four-class output should be treated as an *advisory classification* that guides crew preparation (e.g. internal vs. external leak equipment) rather than as a hard dispatch decision. Utilities that require only binary anomaly detection — the majority of current SCADA deployments — can deploy Layer 1 alone with high confidence; the four-class extension adds diagnostic value at the cost of reduced per-class reliability on minority types.

6.3. Cost-Benefit Estimate for a 50,000-Sensor Utility

Adopting the recipe in a utility with 50,000 deployed sensors, each producing one feature vector per day, yields a daily input volume of 50,000 rows. At < 5 ms per 1,000 rows the daily inference budget is < 0.25 s on a single CPU core; on a single 16-core node the recipe scales easily to several million sensors per day. The training side requires 65 MB peak RAM and zero GPU, making the entire pipeline deployable on the same edge gateways currently running threshold-rule firmware.

6.4. SCADA / IoT Integration Recipe

The following deployment template is recommended:

1. **Edge**: each gateway computes the 57 causal features from the 1 s vibration window and forwards them via MQTT/HTTPS to the central inference layer.
2. **Central**: the hierarchical Stage 1 binary gate processes incoming feature vectors at < 5 ms/1k rows; only positive Stage 1 verdicts are forwarded to Stage 2.
3. **Decision**: Stage 2 produces a 4-class probability vector, which is logged to SCADA historian and gated through a per-region operational threshold to dispatch field investigation.
4. **Audit**: every dispatched alarm is followed by a field-confirmed label, which is appended to the active-learning queue for next-quarter retraining.

6.5. Portability to Other Sensor Benchmarks

The methodology of Sections 3.3–3.5 is benchmark-agnostic. The applicability checklist of Section 3.11 identifies the pre-conditions a candidate sensor benchmark must satisfy: (i) an identifier-like column (sensor ID, patient ID, device ID, session ID); (ii) a natural group hierarchy (sensor / site / region / time); (iii) features that group physically (frequency bands, channels, trajectory segments); and (iv) a published row-level CV result that the audit can target. Table 15 maps the methodology to four representative external benchmarks and identifies how each probe would instantiate.

Three of these portability targets (UCI SECOM McCann and Johnston (2008), UCI HAR Anguita et al. (2013), and the seven machines of MIMII-DG Dohi et al. (2022)) are directly evaluated in Section 5.8 of this paper; the

Table 15: Portability of the four-probe shortcut audit and three-tier OOD ladder to four representative sensor-ML benchmarks. The audit instantiation is benchmark-specific ("identifier" column varies); the ladder instantiation follows from the available group hierarchy. External validation of the methodology on these benchmarks is identified as the immediate future-work priority.

Benchmark	Sensor / Task	Identifier (P1)	Group hierarchy (P3)	Physical feature group
AI Hub (this paper)	Pipe vibration / leak typing	`sid`	*si* / *si-gun-gu* / *gwangyeoksi-do*	512 frequency bands
MIMII Koh et al. (2021)	Industrial machine sound	machine ID	machine type / location	log-mel spectrogram
ToyADMOS	Toy machine sound	ToyCar/ToyConv. ID	model variant / time	log-mel spectrogram
SECOM (UCI)	Semiconductor process	wafer/lot ID	process tool / shift	sensor channels
EEG-MMIDB	Motor-imagery EEG	subject ID	subject / session / electrode	64 EEG channels

remaining audio targets (ToyADMOS-DG and any utility-collected internal vibration corpus) are identified as follow-up technical notes. The present paper releases the benchmark-agnostic audit, OOD ladder, and grouped permutation-importance harness as a reusable utility so that practitioners can apply the same diagnostics to their own datasets without re-implementing them.

6.6. Practitioner Recommendations

Five concrete actions for practitioners considering the AI Hub pipeline (or any vibration-sensor leak benchmark) for field deployment:

1. **Run the four-probe shortcut audit first.** If $(\epsilon, \delta) \approx (0, 0)$, the benchmark headline is identifier-driven and downstream architecture search is misallocated effort.
2. **Drop identifier columns during training.** On AI Hub, removing `site`, `sid`, `lrate`, `llevel`, and the regional codes provides zero accuracy benefit and slightly improves macro-F1 under group-CV.
3. **Restrict to 65–422 Hz before training the 4-class typer.** The 57-feature subset trains $\sim 10\times$ faster than the full 534-feature set and improves OOD macro-F1.
4. **Deploy two stages, not one.** Use the binary energy gate for alarms and consult the 4-class typer only when the gate fires.
5. **Report LORO, not row-CV.** Row-level CV is a useful internal debugging tool, but the number that matters for field decisions is LORO-weighted macro-F1 on the most class-balanced region.

6.7. Compliance and Standards

The shortcut audit and three-tier OOD protocol align with the NIST AI RMF National Institute of Standards and Technology (2023) "Govern — Map — Measure" core functions and with the EU AI Act European Parliament and Council (2024) high-risk-classification disclosure requirements for infrastructure AI. The audit produces an evidence trail (`claims.csv`, multi-seed mean ± std, fold-level JSONs) that supports both regulatory regimes.

7. Threats to Validity and Failure Modes

Single leak-detection benchmark. The 5-class *leak typing* results are restricted to the AI Hub vibration dataset; replication on an independent vibration-leak benchmark (e.g. a utility-collected internal dataset or the audio-spectrogram MIMII-DG release) is the natural next step for the deployment-recipe claim and is identified as ongoing future work. The methodological claims (four-probe audit, three-tier OOD ladder) have been replicated on nine independent external benchmarks (SECOM McCann and Johnston (2008), UCI HAR Anguita et al. (2013), and the seven machines of MIMII-DG Dohi et al. (2022); Section 5.8, Table 13), which establishes the *protocol's* generality even though the *leak-typing accuracy* is benchmark-specific.

Korean-specific deployment and pipe material. Sensors, pipe materials, and water pressures in Korean networks may differ from European or North-American counterparts. The 121–422 Hz band finding is broadly consistent with published acoustics for plastic pipes Hunaidi et al. (2000); Almeida et al. (2014), but the precise band edges depend on pipe material and diameter: metal pipes typically conduct leak noise at higher frequencies than plastic Hunaidi et al. (2000). The AI Hub metadata does not include per-sensor pipe-material labels, so a material- stratified analysis

is not possible with the current dataset. Future work with material-labelled utility data should re-verify the band edges per material type.

Multi-seed coverage. Mean ± std across five seeds is reported for every stochastic experiment. Tree experiments are deterministic given the seed and feature ordering; deep experiments inherit non-determinism from CUDA (cudnn benchmark mode disabled).

Sensor drift and seasonality. The dataset spans roughly one year; multi-year drift (sensor ageing, pipe-material fatigue, seasonal water-table changes) is not characterised. A chronological 70/30 temporal split (train on recordings before November 2020, test on recordings after) is reported in Table 14 as a complementary validation axis to the spatial LORO split. Longer-horizon temporal validation requires a multi-year data collection that is not currently available.

Adversarial deployment shift. Active sabotage of sensor placement (loose mounts, off-axis installation) is outside the scope of this evaluation.

Single-author evaluation. As a single-author study, the manuscript has not undergone internal multi-reviewer scrutiny before submission. All numeric claims are anchored to result JSONs (see `claims.csv` in the release), and the integrity audit listed in the supplementary material describes the cross-checks performed; nonetheless, external peer review remains the most reliable correction mechanism, and the released code/data package is designed to support immediate replication.

Class imbalance on the canonical fold. The Chungbuk LORO fold contains only 45 `normal` samples versus 5,029 leak samples (98.1 % positive base rate), which makes the Stage-1 binary detector look trivially strong on this fold alone. The cross-LORO calibrated-FAR analysis (Section 5.6, Table 11) ameliorates but does not eliminate this concern: only 4 of 11 LORO folds contain ≥ 50 normal samples, so the operational FAR estimate is itself noisy on the smaller folds and should be cross-validated on a utility-collected internal benchmark before field promotion.

Domain-knowledge prior. The 121–422 Hz finding matches published acoustics for plastic pipes Hunaidi et al. (2000); Almeida et al. (2014); Gao et al. (2017), so the "discovery" is at least partially a *re-discovery*. The contribution here is the audited methodology that recovers a nameable physical band from the data without supervision, not the band itself; in benchmarks where no such prior exists the methodology would still apply but the result would be harder to corroborate against external evidence.

Hierarchical Stage-1 simplicity. The current Stage-1 classifier is a single-call binary XGBoost. More elaborate designs (e.g. a calibrated ensemble of Stage-1 predictors, or a Bayesian-fusion gate that integrates the abnormal-typer confidence) could plausibly improve operational accuracy but were deliberately excluded to keep the deployment recipe auditable and reproducible on edge hardware.

8. Conclusions and Future Work

This paper provided a field-deployment-oriented evaluation of machine-learning water-leak detection on the AI Hub vibration benchmark, to the author's knowledge the first study to apply sensor-, site-, and region-group holdouts to this dataset together with a formal shortcut-audit protocol and a multi-seed harness. Four findings stand out. (1) The published 99.79 % accuracy is driven by identifier-label couplings; a four-rule lookup that ignores the vibration spectrum reproduces 99.77 %. (2) Under LORO, the same XGBoost falls to weighted accuracy 56.8 % and macro-F1 0.304. (3) Three independent attribution methods converge on 121–422 Hz as the robustly generalisable leak band (termed "causal" in the operational OOD sense; Remark 1), and 422–1470 Hz is exposed as an active confounder. (4) A pre-specified tree-only soft average of flat and hierarchical causal XGBoost classifiers attains OOD macro-F1 **0.4326** with perfect normal-class recall, deployed at < 5 ms per 1,000 rows on CPU; LightGBM achieves a deterministic 0.4341 at the cost of 127× longer training. The audit protocol is validated on nine external sensor benchmarks spanning three modalities, and a temporal-split experiment (Table 14) confirms that the 121–422 Hz band generalises across both space and time within the dataset's one-year window.

Future work.. Six directions are foreseen: (a) replicate the audit on an independent vibration-leak dataset to test the 121–422 Hz finding's transferability; (b) train a self-supervised spectrogram backbone on the unlabelled recordings to close the deep-vs-tree gap on `normal`-class recall; (c) extend the threshold sweep with utility-side cost asymmetries (a missed leak costs more than a false alarm, by an empirically estimable factor); (d) apply the four-probe audit to other Korean smart-water benchmarks distributed by AI Hub, K-water K-water (Korea Water Resources Corporation)

(2022), and KICT; (e) a randomised field experiment that physically masks specific frequency bands at recording time would establish an interventionist causal claim for the 121–422 Hz band in the structural-causal-model sense (Remark 1); (f) a multi-year longitudinal data collection enabling temporal drift characterisation beyond the one-year window analysed here (Table 14).

CRediT Authorship Contribution Statement

Suan Lee: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration, Funding acquisition. (Single-author submission; the author is solely responsible for all CRediT roles applicable to this work.)

Declaration of Competing Interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research was supported by Semyung University (grant number 2025).

Data Availability

The AI Hub water-pipe leak vibration dataset is openly distributed by NIA Korea AI Hub NIA Korea AI Hub (2022) (dataset ID 138, accessed on 5 October 2023) under the Korean Public Data Open Licence Type 1, and is freely downloadable from `https://www.aihub.or.kr/`. The full experimental harness, multi-seed runner, 30+ result JSON files, fold-level logs, and figure-generation scripts that reproduce every numeric claim in this paper are released under the MIT licence at `https://github.com/suanlab/causal-leak`.

Acknowledgements

The author thanks the AI Hub curators for maintaining and publicly distributing the water-pipe vibration dataset.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this manuscript, the author used a large-language-model assistant (Claude, Anthropic) to support LaTeX formatting, English copy-editing, and Python scripting for figure generation and result aggregation. All experimental design, methodology, data analysis, interpretation of results, and scientific conclusions are solely the author's own. After using this tool/service, the author reviewed and edited all content and takes full responsibility for the content of the publication.

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